# MA 140 : MATHEMATICAL ANALYSIS 1

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## Acknowledgements

These notes are an adaptation of James Robinson's MA140 notes from Autumn 2024.

## Motivation

The aim of this course is to build the mathematical tools that we need to answer (reasonably well) the following questions:

(1) What does it mean to say that $0.9999\cdots = 1$?

(2) What does it mean when we write that

$$\left(1+\frac{1}{n}\right)^n \to e$$

as $n \to \infty$?

(3) In fact, how do we rigorously define $e$? What does it mean when we write

$$\sum_{n=1}^{\infty} \frac{1}{n!}?$$

More generally, when can we add infinitely many things?

(4) What do we mean when we call a function continuous? *"You can draw it without taking your pencil off the paper"* is not a definition suitable enough for writing mathematical proofs. Also, what about functions which are somewhat painful to draw? For instance, let

$$f(x) = \begin{cases} 1/q, & \text{when } x = p/q \text{ for some co-prime } p, q \in \mathbb{N} \\ 0, & \text{for all other } x \in \mathbb{R}. \end{cases}$$

## Preliminaries

We will use $\mathbb{N}$ to denote the natural numbers

$$\mathbb{N} = \{1, 2, 3, 4, \dots, \},$$

$\mathbb{Z}$ to denote the integers

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\},$$

$\mathbb{Q}$ to denote the rational numbers

$$\mathbb{Q} = \{p/q : p \in \mathbb{Z}, q \in \mathbb{N}\},$$

and $\mathbb{R}$ to denote the set of all real numbers. We will use $\mathbb{R} \setminus \mathbb{Q}$ to denote the irrational numbers, that is, $\mathbb{R} \setminus \mathbb{Q}$ is the set of all $x \in \mathbb{R}$ such that $x \notin \mathbb{Q}$.

*Remark* 0.1. We will, at this point, not give a precise definition of what the real numbers are or how these are constructed.

From this point onwards, you can assume that the set $\mathbb{R}$ of real numbers is a set containing the rational numbers. You can add, subtract and multiply real numbers, and this matches addition, subtraction and multiplication for rationals when the said real numbers are restricted to rational numbers. Same comment holds for division when the divisor is non-zero. These operations interact with each other in the usual sense: for any $a, b, c \in \mathbb{R}$, we have

$$2(a + b - c) = 2a + 2(b - c) = 2a - 2c + 2b \quad \text{and} \quad (4a)/7 = 4(a/7) = a(4/7).$$

Moreover, you can assume that the real numbers are ordered in the usual sense: for any $x, y \in \mathbb{R}$, we have either $x < y$, or $x > y$, or $x = y$. Again this ordering, when restricted to rational numbers, matches the standard notion of order over rationals. You can also assume that arithmetic and order over reals interact in the standard way that you know: if $a < b$, then $3.14a < 3.14b$ and $a + 10 < b + 10$.

For a more precise description of these assumptions, see §5.1, and for a handwavy construction of the real numbers, see §5; although both these sections are non-examinable.

*What can we not assume?* Functions like log, sin, cos or existence of numbers such as $\pi, e$ or even $34^{1/100}$. One of the aims of this module, and subsequent modules in analysis, is to develop a concrete notion of what these numbers and functions mean and how to define them properly.

**Notation 0.2.** Here is some useful notation.

(1) We will use Greek letters: $\alpha$ (*alpha*), $\beta$ (*beta*), $\varepsilon$ (*epsilon*), $\delta$ (*delta*). Occasionally, we might also use $\eta$ (*eta*), $\sigma$ (*sigma*), $\gamma$ (*gamma*), $\theta$ (*theta*). We will (most probably) not use $\chi$ (*chi*), $\nu$ (*nu*), $\mu$ (*mu*), $\lambda$ (*lambda*).

(2) Intervals are sets of real numbers written as

$$(-\infty, b), (-\infty, b], (a, b), (a, b], [a, b), [a, b], (a, \infty), [a, \infty), \text{ or } \mathbb{R},$$

where $a, b$ are real numbers. We use a round bracket for a strict inequality and a square bracket for a non-strict inequality, for example,

$$(a, b] = \{x \in \mathbb{R} : a < x \leqslant b\}.$$

Intervals of the form $(a, b)$ are called *open* and intervals of the form $[a, b]$ are called *closed*.

(3) Some useful mathematical shorthand symbols: '$\exists$' means 'there exists/exist' and '$\forall$' means 'for all/every'. Thus the statement: '*for every* $n \in \mathbb{N}$, *there exists* $m \in \mathbb{Q}$ *such that* $n < m$' can be rewritten as '$\forall\ n \in \mathbb{N},\ \exists\ m \in \mathbb{Q}$ *such that* $n < m$'.

(4) Given positive integer $r$ and real numbers $x_1, \ldots, x_r$ (which may not be distinct), we define

$$\min\{x_1, \ldots, x_r\}$$

to be some $x_i$ satisfying $x_i \leqslant x_j$ for all $1 \leqslant j \leqslant r$. We similarly define $\max\{x_1, \ldots, x_r\}$ to be some $x_k$ satisfying $x_k \geqslant x_j$ for all $1 \leqslant j \leqslant r$. Moreover, given a set $S = \{s_1, \ldots, s_r\}$ of real numbers and some function $f : S \to \mathbb{R}$, we define

$$\min_{s \in S} f(s) = \min\{f(s_1), \ldots, f(s_r)\} \quad \text{and} \quad \max_{s \in S} f(s) = \max\{f(s_1), \ldots, f(s_r)\}. \tag{0.1}$$

The following is known as Bernoulli's inequality.

**Lemma 0.3.** *For all* $n \in \mathbb{N}$ *and* $x \geqslant 0$, *one has*

$$(1 + x)^n \geqslant 1 + nx.$$

*Proof.* By binomial theorem, we have

$$(1 + x)^n = \sum_{j=0}^{n} \binom{n}{j} x^j, \quad \text{where} \quad \binom{n}{j} = \frac{n!}{(n - j)!j!}.$$

Since all the above terms are non-negative and $n \geqslant 1$, we get that

$$(1+x)^n \geqslant \sum_{j=0}^{1} \binom{n}{j} x^j = 1 + nx. \qquad \square$$

There are other ways to prove this inequality, for example, via induction on $n$. Moreover, this inequality holds for all $x > -1$, see examples sheet.

**Notation 0.4.** Many of the results we prove state a hypothesis A and then draw the conclusion B as a consequence. We have various ways of saying this, and of writing it down:

- if A, then B
- A implies B
- A $\Rightarrow$ B
- B follows from A
- B if A
- B $\Leftarrow$ A.

Suppose that A implies B (A $\Rightarrow$ B) and B implies A (B $\Rightarrow$ A). Then we can say

- A $\Leftrightarrow$ B
- A and B are equivalent
- A if and only if B.

For the last item on this list, note that 'A if B' is the same as 'B $\Rightarrow$ A'. Moreover, 'A only if B' is the same as 'A holds only if B holds', that is, 'if B does not hold, then A does not hold'. This is in fact equivalent to 'A $\Rightarrow$ B' and is known as the 'contrapositive'.

**Lemma 0.5.** *Let $n \in \mathbb{N}$ and let $x, y \in \mathbb{R}$ satisfy $x, y > 0$. Then*

$$\begin{aligned} x < y &\quad\Leftrightarrow\quad x^n < y^n, \\ x \leqslant y &\quad\Leftrightarrow\quad x^n \leqslant y^n, \\ x = y &\quad\Leftrightarrow\quad x^n = y^n. \end{aligned}$$

*Proof.* We will first prove that $x < y \Rightarrow x^n < y^n$.

Thus, suppose $x, y > 0$ satisfy $x < y$. We will prove by induction that $x^k < y^k$ for all $k \in \mathbb{N}$. The hypothesis implies that this statement is true when $k = 1$. Now suppose we have proved this statement for some arbitrary $k \in \mathbb{N}$, that is, we know $x^k < y^k$. Moreover, we have $x < y$. Then note that

$$y^{k+1} - x^{k+1} = y^{k+1} - yx^k + yx^k - x^{k+1} = y(y^k - x^k) + x^k(y - x).$$

We now claim that the left hand side is positive: indeed, $y > 0$ from the hypothesis and $y^k - x^k > 0$ from the inductive hypothesis. Moreover, $x^k > 0$ since $x > 0$, and $y - x > 0$

from the hypothesis. Thus, our claim is true, and so, we have $y^{k+x} > x^{k+1}$. Induction now implies that $x^n < y^n$ for all $n \in \mathbb{N}$.

It is clear that if $x = y$, then $x^n = y^n$ for all $n \in \mathbb{N}$. This combines with the preceding statement to tell us that $x \leqslant y \Rightarrow x^n \leqslant y^n$.

We will now prove that $x^n < y^n \Rightarrow x < y$ for all $n \in \mathbb{N}$. We will prove this via contradiction, and so, suppose that $x, y > 0$ satisfy $x^n < y^n$ for some $n \in N$ and that $x \geqslant y$. Using the preceding statement (with roles of $x, y$ reversed), we see that the latter statement tells us that $x^n \geqslant y^n$, but this contradicts our hypothesis that $x^n < y^n$.

The remaining statements '$x^n = y^n \Rightarrow x = y$' and '$x^n \leqslant y^n \Rightarrow x \leqslant y$' can be proved similarly. $\square$

## 1. The real numbers

This chapter outlines various properties of real numbers that match some of our usual intuition but are worth writing formally.

### 1.1. **Rationals are not 'complete'.**

**Proposition 1.1.** *There is no $x \in \mathbb{Q}$ such that $x^2 = 2$.*

*Proof.* We prove this via contradiction. Suppose there is a rational number $x$ such that $x^2 = 2$. Since $x$ is rational, we must have $x = p/q$, where $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. We may further assume that $p/q$ is a fraction in its lowest terms, that is, $p$ and $q$ have no common factors.

The hypothesis $x^2 = 2$ implies that $(p/q)^2 = 2$, that is, $p^2 = 2q^2$. Since the right hand side is divisible by 2, we must have that the left hand side is divisible by 2. We thus deduce that 2 divides $p^2$. This can only happen if 2 divides $p$. Therefore, $p = 2m$ for some $m \in \mathbb{Z}$.

Substituting this in the equation $p^2 = 2q^2$, we get that $4m^2 = 2q^2$, that is, $2m^2 = q^2$. As before, we see that this implies that 2 divides $q^2$, which, in turn, tells us that 2 divides $q$. But now we know that 2 divides both $p$ and $q$, which contradicts our hypothesis that $p$ and $q$ did not have a common factor. □

*Remark** 1.2. There are many other ways to prove Proposition 1.1. Later in the course, we will see a different proof which relies on the fact that if $x \in \mathbb{Q}$ satisfies $x^2 = 2$, then $0 < x - 1 < 1$.

*Lecture 2*

### 1.2. **Upper and lower bounds.**

Our next aim is now to show that there is some $x \in \mathbb{R}$ such that $x^2 = 2$. In order to show this, we will need to introduce a key property of the real numbers, which itself requires some new definitions.

**Definition 1.3.** Suppose that $S$ is a set of real numbers. An *upper bound* for $S$ is a real number $r$ such that $s \leqslant r$ for all $s \in S$. If $S$ has an upper bound, then we say that $S$ is *bounded above*.

Similarly, a *lower bound* for $S$ is a real number $t$ such that $s \geqslant t$ for all $s \in S$. If $S$ has a lower bound, then it is *bounded below*.

If a set is bounded both above and below, then we say that it is *bounded*.

If a set has one upper bound, then it has (infinitely) many: if $s \leqslant r$ for all $s \in S$, then for any $r' > r$, we also have that $s \leqslant r'$ for all $s \in S$.

*Remark* 1.4. Note that $s \leqslant r$ for all $s \in S$ if and only if $-s \geqslant -r$ for all $s \in S$. Writing

$$-S = \{-s : s \in S\},$$

the above means that $S$ is bounded above if and only if $-S$ is bounded below. Moreover, this means that any statement for upper bounds will imply an analogous statement for lower bounds.

**Example 1.5.** Following are some helpful examples.

(1) Given $a, b \in \mathbb{R}$ with $a < b$, the intervals $(a, b)$, $(a, b]$, $[a, b)$, $[a, b]$ are bounded below by $a$ and bounded above by $b$.

(2) The set $S = \{1/n : n \in \mathbb{N}\} = \{1, 1/2, 1/3, \dots\}$ is bounded below by 0 and bounded above by 1. Note that $1 \in S$ but $0 \notin S$.

(3) The interval $(-10, \infty)$ is bounded below by $-10$ but it is not bounded above.

(4) The set $S = \{n/(n+1) : n \in \mathbb{N}\} = \{1/2, 2/3, 3/4, \dots\}$ is bounded above by 2 and bounded below by 0, but neither of these are the 'best' bounds.

(5) We claim that the set $S = \{(1 + 1/n)^n : n \in \mathbb{N}\}$ is bounded below by 2 and bounded above by 3. The lower bound follows from Bernoulli's inequality (Lemma 0.3), indeed, for all $n \in \mathbb{N}$, we have

$$(1 + 1/n)^n \geqslant 1 + (1/n) \cdot n = 2.$$

As for the upper bound, binomial theorem implies that

$$\left(1 + \frac{1}{n}\right)^n = \sum_{j=0}^{n} \binom{n}{j} \frac{1}{n^j} = 1 + \sum_{j=1}^{n} \frac{1}{j!} \frac{n(n-1)\dots(n-j+1)}{n^j} \leqslant 1 + \sum_{j=1}^{n} \frac{1}{j!}. \tag{1.1}$$

Whenever $j \geqslant 2$, we see that $j! \geqslant j(j-1)$. Thus, we have that

$$\begin{aligned}\left(1 + \frac{1}{n}\right)^n &\leqslant 2 + \sum_{j=2}^{n} \frac{1}{j!} \leqslant 2 + \sum_{j=2}^{n} \frac{1}{j(j-1)} \\ &= 2 + \sum_{j=2}^{n} \left(\frac{1}{j-1} - \frac{1}{j}\right) = 2 + 1 - \frac{1}{n} < 3.\end{aligned}$$

This proves that 3 is an upper bound for $S$. Are these the 'best' lower and upper bounds for $S$?

(6) Finally, what about the set $\mathbb{N}$ itself? This is lower bounded by 1. Does it have an upper bound?

1.3. **Least upper bounds.** In order to answer the question raised in (6), we will need an additional property of $\mathbb{R}$.

**Definition 1.6.** A real number $l$ is a *least upper bound* for a set $S$ if $l$ is an upper bound for $S$ and if for any upper bound $L$ of $S$, we have $l \leqslant L$. One can similarly define a *greatest lower bound* for the set $S$.

If $l \in \mathbb{R}$ is a least upper bound for some set $S$, then we say that $l$ is the *supremum* of $S$, or equivalently, $l = \sup S$. Similarly, if $m$ is the greatest lower bound for some set $S$, then we say that $m$ is the *infimum* of $S$ or $m = \inf S$.

**Lemma 1.7.** *Any set of real numbers can have at most one least upper bound.*

*Proof.* In order to see this, suppose a set $S$ of real numbers has two least upper bounds $l_1$ and $l_2$. Since $l_1$ is a least upper bound for $S$, it must be an upper bound for $S$. Moreover, since $l_2$ is a least upper bound for $S$ and $l_1$ is an upper bound for $S$, then we must have $l_2 \leqslant l_1$. Similarly, we can show that $l_1 \leqslant l_2$. This implies that $l_1 = l_2$. □

*Lecture 3*

We now introduce the additional axiom that makes the real numbers behave as we want; it is one version of the so-called "Completeness Axiom" for the real numbers.

**Least Upper Bound Axiom.** *Any non-empty set of real numbers that is bounded above has a least upper bound.*

This is equivalent to the statement that any non-empty set of real numbers that is bounded below must have a greatest lower bound, see Remark 1.4.

**Lemma 1.8.** *Let $l$ be a real number and let $S \subset \mathbb{R}$ be some non-empty set. Then $l = \sup S$ if and only if $l$ is an upper bound for $S$ and for all every $t < l$, there exists some $s \in S$ such that $t < s$.*

*Proof.* Let $l$ be an upper bound for the set $S$. It suffices to show that the following statements are equivalent:

(1) For any upper bound $L$ of $S$, we have $l \leqslant L$
(2) For any $t < l$, there exists some $s \in S$ such that $t < s$.

We will first show that (1) $\Rightarrow$ (2), and we will do so via contradiction. Suppose $t \in \mathbb{R}$ satisfies $t < l$. For purposes of contradiction, we assume that $t \geqslant s$ for all $s \in S$. This means that $t$ is an upper bound for $S$. By property (1), we get that $l \leqslant t$, contradicting our assumption that $t < l$.

We will now show that (2) $\Rightarrow$ (1), again by contradiction. Thus, suppose that $L$ is an upper bound of $S$ and suppose for contradiction that $L < l$. Applying (2), this must mean that there exists some $s \in S$ such that $L < s$, contradicting the fact that $L$ was an upper bound of $S$. $\square$

Lemma 1.8 is a formal version of the intuitive statement that a least upper bound of some non-empty set $S \subset \mathbb{R}$ is, in some sense, the 'best' upper bound of $S$.

*Remark* 1.9. Lemma 1.8 may be rewritten as '$l = \sup S$ if and only if for every $\varepsilon > 0$, there exists some $s \in S$ such that $l - \varepsilon < s \leqslant l$.'

Moreover, Lemma 1.8 implies that '$m = \inf S$ if and only if $m$ is a lower bound of $S$ and for any $t > m$, one can find some $s \in S$ such that $m \leqslant s < t$'. In order to see this, use the trick from Remark 1.4.

**Example 1.10.** We revisit some of the sets we described in Example 1.5.

(1) Let $a < b$ be real numbers. Then

$$\sup(a, b) = \sup(a, b] = \sup[a, b) = \sup[a, b] = b$$

and

$$\inf(a, b) = \inf(a, b] = \inf[a, b) = \inf[a, b] = a$$

We will prove that $\sup(a, b) = b$, the rest of the cases follow similarly.

By definition, $b$ is an upper bound for $(a, b)$. Moreover, suppose $t \in \mathbb{R}$ satisfies $t < b$. If $t \leqslant a$, then we are done since $t < (a + b)/2$ and $(a + b)/2 \in S$. On the other hand, if $a < t < b$, then consider the element $(b + t)/2$. We have $(b + t)/2 \in (a, b)$ and $t < (b + t)/2 < b$. Thus by Lemma 1.8, we see that $b = \sup(a, b)$.

This also tells us that the supremum of a set, if it exists, may not lie in the set itself.

(2) Let $S = \{1/n : n \in \mathbb{N}\}$. We claim that $\sup S = 1$. Indeed, 1 is an upper bound for $S$, and moreover, for any $t < 1$, we have $t < 1$ with $1 \in S$. Applying Lemma 1.8 now gives us the desired conclusion. What about $\inf S$?

(3) Let $S = (-10, \infty)$. Then $-10 = \inf S$ and $S$ does not have a least upper bound since it is not bounded above.

(4) Let $S = \{n/(n+1) : n \in \mathbb{N}\}$. As in example (2), we can prove that $\inf S = 1/2$. What about $\sup S$?

(5) Let $S = \{(1+1/n)^n : n \in \mathbb{N}\}$. We proved that 2 is a lower bound for $S$. Moreover, since $2 \in S$, we get that $2 = \inf S$. What about $\sup S$? It exists since $S$ is non-empty and bounded above by 3.

(6) Let $S = \mathbb{N}$. We see that $\inf \mathbb{N} = 1$. Is $\mathbb{N}$ bounded above?

Let us focus on the set described in 2, that is, $S = \{1/n : n \in \mathbb{N}\}$. We know that $\inf S$ exists since it is non-empty and bounded below by 0. In fact, we will show, using the Least upper bound axiom that $0 = \inf S$. Noting Lemma 1.4, this is equivalent to showing that for any real number $m > 0$, there exists some $n \in \mathbb{N}$ such that $1/n < m$.

**Archimedean Principle.** *For any real number $r$, there exists $n \in \mathbb{N}$ such that $r < n$.*

*Proof.* Let $r$ be a real number. If Archimedean principle does not hold for this $r$, then this means that for all $n \in \mathbb{N}$, we have $r \geqslant n$. Thus, $r$ is an upper bound for the set $\mathbb{N}$ of natural numbers. Since $\mathbb{N}$ is non-empty, the Least upper bound axiom implies that $\sup \mathbb{N}$ exists. Let $l = \sup \mathbb{N}$. We see that $l$ is an upper bound for $\mathbb{N}$, whence, $l \geqslant n+1$ for all $n \in \mathbb{N}$. In particular, this means that $l - 1 \geqslant n$ for all $n \in \mathbb{N}$, that is, $l - 1$ is an upper bound for $\mathbb{N}$. Since $l = \sup \mathbb{N}$, we may apply Lemma 1.8 to deduce that $l \leqslant l - 1$, which delivers the desired contradiction. □

Archimedean principle immediately implies that for any $m > 0$, there must exist some $n \in \mathbb{N}$ such that $1/n < m$. Thus, we have $\inf\{1/n : n \in \mathbb{N}\} = 0$. We can now similarly analyse the situation described in (4) and show that

$$\sup\{n/(n+1) : n \in \mathbb{N}\} = 1.$$

Finally, Archimedean principle tells us that $\mathbb{N}$ is not bounded above, answering the question raised in (6).

*Lecture 4*

1.4. **A real solution to $x^2 = 2$.** Using the Least upper bound axiom, we will now show that there exists some real number $x$ such that $x^2 = 2$.

**Proposition 1.11.** *There exists a unique real number $l > 0$ such that $l^2 = 2$.*

*Proof.* Let

$$S = \{x \in \mathbb{R} : x \geqslant 0 \text{ and } x^2 < 2\}.$$

Note that $1 \in S$, and so, the set $S$ is non-empty. Moreover, the set $S$ is bounded above. Indeed, if $x \in S$, then $x \geqslant 0$ and $x^2 < 2 < 2^2$, in which case, we may apply Lemma 0.5 to get that $x < 2$. We can now apply the Least upper bound axiom to deduce that $\sup S$ exists.

Thus, let $l = \sup S$. Note that $l > 0$ since $1 \in S$. We will now prove that $l^2 = 2$, and we will do so via contradiction. Hence, suppose that $l^2 \neq 2$, in which case, either $l^2 > 2$ or $l^2 < 2$.

Suppose the first case holds, that is $l^2 > 2$. In this case, we will show that there exists some $\varepsilon > 0$ such that $l - \varepsilon$ is an upper bound for $S$, which will contradict the definition of $l$ being the least upper bound of $S$. In order to find this $\varepsilon$, note that

$$(l - \varepsilon)^2 = l^2 - 2l\varepsilon + \varepsilon^2 \geqslant l^2 - 2l\varepsilon. \tag{1.2}$$

Setting

$$\varepsilon = \min\left\{\frac{l^2 - 2}{4l}, \frac{l}{2}\right\},$$

we see that $l - \varepsilon > 0$. Moreover, (1.2) implies that

$$(l - \varepsilon)^2 - 2 \geqslant l^2 - 2 - 2l\varepsilon \geqslant \frac{2}{l^2 - 2} > 0.$$

Thus, for any $s \in S$, we have $(l - \varepsilon)^2 > 2 > s^2$. We may now apply Lemma 0.5 to deduce that $l - \varepsilon > s$ for all $s \in S$, and so, $l - \varepsilon$ is an upper bound for $S$. This delivers the required contradiction.

We now assume that we are in the second case, that is, $l^2 < 2$. In this case, we will show that there exists some $\delta > 0$ such that $l + \delta \in S$, which will contradict the fact that $l$ is an upper bound of $S$. In fact, we will choose

$$\delta = \min\left\{\frac{2 - l^2}{6l}, \frac{l}{2}\right\}.$$

This choice of $\delta$ ensures that

$$2 - (l + \delta)^2 = (2 - l^2) - 2\delta l - \delta^2 > (2 - l^2) - 3\delta l \geqslant (2 - l^2) - \frac{2 - l^2}{2} > 0.$$

This means that $(l + \delta)^2 < 2$. Moreover, since $l + \delta > 0$, we must have $l + \delta \in S$, which provides the required contradiction.

Thus, we have shown that $l^2 = 2$. We will now show that if $x > 0$ satisfies $x^2 = 2$, then $x = l$. This, in fact, immediately follows from Lemma 0.5. □

*Remark* 1.12. We can similarly show that for any $q \in \mathbb{N}$ and any real number $y > 0$, there exists a unique real number $x > 0$ such that $x^q = y$.

For any $y > 0$ and any $q \in \mathbb{N}$, we will refer to the unique positive real number $x$ satisfying $x^q = y$ as $y^{1/q}$. Moreover, given any $p \in \mathbb{Z}$ and any $q \in \mathbb{N}$, we now define

$$y^{p/q} = (y^p)^{1/q}.$$

Thus, we can now define real numbers like $3^{1/5}$ or $80^{81/80}$. On the other hand, what is $2^{2^{1/2}}$?

*Lecture 5*

1.5. **Rationals and irrational numbers are interleaved.** We have shown that there exists at least one irrational number $2^{1/2}$, that is, $\mathbb{R} \setminus \mathbb{Q}$ is non-empty. In fact, there are infinitely many such numbers, indeed, if $\alpha$ is irrational and $p/q \in \mathbb{Q}$, then $\alpha p/q$ and $\alpha + p/q$ are also irrational. Indeed, if not, then we would have

$$\alpha + p/q = p'/q', \quad \text{that is,} \quad \alpha = (p'q - pq')/qq' \in \mathbb{Q},$$

which gives us a contradiction. A similar argument shows that $\alpha p/q \in \mathbb{R} \setminus \mathbb{Q}$.

It turns out that rational and irrational numbers are interleaved. We will first show that every real number has a unique maximal integer $n$ such that $n \leqslant x$.

**Lemma 1.13.** *For every $x \in \mathbb{R}$, there exists a unique $n \in \mathbb{Z}$ such that $n \leqslant x < n + 1$.*

*Proof.* Let $S = \{k \in \mathbb{Z} : k \leqslant x\}$. This is non-empty (indeed, apply Archimedean principle to find some $l \in \mathbb{Z}$ such that $l > -x$) and this is bounded above by $x$. Thus let $r = \sup S$. Noting Remark 1.9, we may find some $n \in S$ such that $r - 1/2 < n \leqslant r$.

Note that $n \leqslant x$ since $n \in S$. Moreover, since $n + 1 > n + 1/2 > r$ and $r = \sup S$, we see that $n + 1 \notin S$, and so, $n + 1 > x$.

Such an $n \in \mathbb{Z}$ must be unique. Indeed, suppose $n < m \in \mathbb{Z}$ satisfy $n \leqslant x < n + 1$ and $m \leqslant x < m + 1$. Since $m > n$, we must have $m \geqslant n + 1$, which implies that $m \geqslant n + 1 > x$, which contradicts the fact that $m \leqslant x$. □

**Definition 1.14.** For every $x \in \mathbb{R}$, let $\lfloor x \rfloor = n$, where $n$ is the unique integer satisfying $n \leqslant x < n + 1$.

**Lemma 1.15.** *Let $a < b$ be real numbers. Then there exists $y \in \mathbb{Q}$ and $z \in \mathbb{R} \setminus \mathbb{Q}$ such that $y, z \in (a, b)$.*

*Proof.* We will first show that it suffices to show that there exists a rational number $y \in (a, b)$ for every $a < b$. Indeed, if we have this result, then we know that there exists a rational

number $y' \in (a/2^{1/2}, b/2^{1/2})$, that is, $a < 2^{1/2}y' < b$. Since $y' \in \mathbb{Q}$, we may set $z = 2^{1/2}y'$ and note that $z \in (a, b)$ and $z \in \mathbb{R} \setminus \mathbb{Q}$.

We will now show that $y \in \mathbb{Q}$ such that $y \in (a, b)$. Let $x = b - a$. Since $x > 0$, the Archimedean principle implies that there exists some $n \in \mathbb{N}$ such that $n > 10/x$, that is, $nb - na > 10$. It suffices to prove that there exists some $m \in \mathbb{Z}$ such that $m \in (na, nb)$. Indeed, given such an $m \in \mathbb{Z}$, we may set $y = m/n$ and note that $y \in (a, b)$.

In order to prove our claim, we set $m = \lfloor na \rfloor + 1$. By definition, we have $m \in \mathbb{Z}$ and $m > na$. Moreover, we claim that $m < nb$. If not, then $m \geqslant nb > na + 10$, that is, $\lfloor na \rfloor > na+9$. But by definition, we have $\lfloor na \rfloor \leqslant na$, giving us the desired contradiction. □

This lemma in fact tells us that rationals are 'dense' in $\mathbb{R}$, that is, given any real number $x$, we can find some rational number arbitrarily close to $x$.

**Corollary 1.16.** *Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. Then there exists some $y \in \mathbb{Q}$ such that $y \in (x - \varepsilon, x + \varepsilon)$.*

*Proof.* Apply Lemma 1.15 with $a = x - \varepsilon$ and $b = x + \varepsilon$. □

*Lecture 6*

## 1.6. **Absolute value and the triangle Inequality.**

**Definition 1.17.** For any $x \in \mathbb{R}$, we define $|x| = \max\{x, -x\}$. Thus $|x| = x$ if $x \geqslant 0$, and $|x| = -x$ if $x < 0$. We refer to $|x|$ as the *absolute value* of $x$.

We list some basic properties of the absolute value.

**Lemma 1.18.** *Let $x, y \in \mathbb{R}$. Then we have*

*(1) $|x| \geqslant 0$,*
*(2) $|x| = |-x|$,*
*(3) $|xy| = |x||y|$, and*
*(4) if $y \neq 0$, then $|x/y| = |x|/|y|$.*

*Proof.* We divide the proof as follows.

(1) Note that (1) follows from the fact that for any $x \in \mathbb{R}$, either $x \geqslant 0$ or $-x > 0$.

(2) Moreover (2) follows from the definition of absolute value.

(3) One can prove (3) by a case-by-case analysis. Firstly, if $x = 0$, then $xy = 0$, in which case, we have $|xy| = 0 = |x| = |x||y|$. A similar analysis holds when $y = 0$, and so, we may assume that $x, y \neq 0$. Now if $x > 0, y > 0$, then $|xy| = xy = |x||y|$. If $x > 0$ and $y < 0$, we get that $xy < 0$, and so, $|xy| = -xy = x(-y) = |x||y|$. A similar analysis holds for the case when $x < 0$ but $y > 0$. Finally, if $x, y < 0$, then $xy > 0$, and so, $|xy| = xy = (-x)(-y) = |x||y|$.

(4) We now prove (4). Since $y \neq 0$, we see that $1 = y \cdot (1/y)$. We now apply (3) to deduce that

$$1 = |1| = |y \cdot (1/y)| = |y||1/y|.$$

This gives us $|1/y| = 1/|y|$. We can now apply (3) again to get

$$|x/y| = |x \cdot (1/y)| = |x||1/y| = |x|/|y|. \qquad \square$$

The absolute value function allows us to measure distance. For instance, note that for any $r > 0$, we have

$$|x| < r \quad \Leftrightarrow \quad -r < x < r. \tag{1.3}$$

Indeed, if $|x| < r$ and $x \geqslant 0$, then $x < r$; while if $|x| < r$ and $x < 0$, then we must have $-x < r$, that is, $x > -r$. Similarly, if $-r < x < r$, then $-r < -x < r$, and so, $-r < |x| < r$. Since absolute value is always non-negative, we get that $|x| < r$.

Given $a, x \in \mathbb{R}$, the quantity $|x - a|$ measures the distance between $x$ and $a$. Indeed,

$$|x - a| < \varepsilon \quad \Leftrightarrow \quad -\varepsilon < x - a < \varepsilon \quad \Leftrightarrow \quad a - \varepsilon < x < a + \varepsilon \quad \Leftrightarrow \quad x \in (a - \varepsilon, a + \varepsilon).$$

**Lemma 1.19.** *For any $a, b \in \mathbb{R}$, one has the triangle inequality*

$$|a + b| \leqslant |a| + |b|. \tag{1.4}$$

*One also has the reverse triangle inequality*

$$||a| - |b|| \leqslant |a - b|. \tag{1.5}$$

*Proof.* We will first prove (1.4). For any $a \in \mathbb{R}$, we have $-|a| \leqslant a \leqslant |a|$. Similarly, we have $-|b| \leqslant b \leqslant |b|$. Combining the two inequalities gives us

$$-(|a| + |b|) \leqslant a + b \leqslant |a| + |b|.$$

Noting (1.3), we get that $|a + b| \leqslant |a| + |b|$.

We will now prove (1.5). As before, noting (1.3), it suffices to prove that

$$-|a - b| \leqslant |a| - |b| \leqslant |a - b|. \tag{1.6}$$

The first inequality above is equivalent to showing that $|b| \leqslant |a| + |a-b| = |a| + |b-a|$. But this follows from the triangle inequality since $b = a + (b-a)$. Similarly, the second inequality in (1.6) is equivalent to showing that $|a| \leqslant |a-b| + |b|$, which again follows from the triangle inequality since $a = (a-b) + b$. □

Triangle inequality is quite useful: for example suppose we have some real number $x \neq 0$. Then for all $y \in \mathbb{R}$ with $|y - x| < |x|/2$, one has $y \neq 0$. Let's try to prove this. We want to show that $y \neq 0$, that is, $|y| > 0$. But note that $x = (x-y) + y$, and so, triangle inequality tells us that

$$|x| \leqslant |x-y| + |y| = |y-x| + |y| < |x|/2 + |y|.$$

Thus we have $|y| > |x|/2$. Moreover, since $x \neq 0$, we get that $|x| > 0$, and so, $|y| > 0$ as well.

Similarly, given $x, y, z \in \mathbb{R}$, since $x - z = (x-y) + (y-z)$, we can apply the triangle inequality to get that

$$|x-z| \leqslant |x-y| + |y-z|.$$

*Remark** 1.20. One can now see why this is called the triangle inequality. Given a point $\boldsymbol{u} \in \mathbb{R}^2$, let $u_1, u_2$ be the unique real numbers satisfying $\boldsymbol{u} = (u_1, u_2)$. We now define $\|\boldsymbol{u}\| = (u_1^2 + u_2^2)^{1/2}$. Then for any $\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^2$, we have that

$$\|\boldsymbol{x} - \boldsymbol{z}\| \leqslant \|\boldsymbol{x} - \boldsymbol{y}\| + \|\boldsymbol{y} - \boldsymbol{z}\|.$$

This can be interpreted as the familiar fact that given a triangle, the size of one of the sides is always bounded above by the sum of the sizes of the two other sides.

Does this inequality still hold if we instead define $\|\boldsymbol{u}\| = (|u_1|^3 + |u_2|^3)^{1/3}$? What about the case when $\|\boldsymbol{u}\| = \max\{|u_1|, |u_2|\}$?

## 2. Sequences and convergence

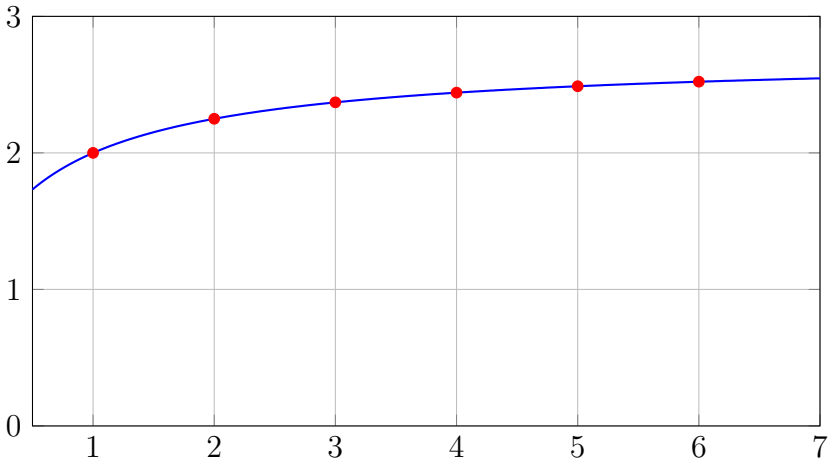
**Definition 2.1.** A sequence $(a_n)_{n=1}^{\infty}$ is an infinite ordered list of real numbers: $a_1, a_2, a_3, a_4, \ldots$. We may also write a sequence as $(a_n)$.

**Example 2.2.** Here are some examples of sequences.

(1) Sequences can be specified by an explicit formula such as $a_n = 1/n$ for all $n \in \mathbb{N}$, or $a_n = (1 + 1/n)^n$ for all $n \in \mathbb{N}$, or $a_n = 2^n - 5^{1/2}$ for all $n \in \mathbb{N}$. The second example looks like:

$$2, \frac{9}{4}, \frac{64}{27}, \frac{625}{256}, \frac{7776}{3125}, \frac{117649}{46656}, \ldots$$

(2) They can be described by plotting points on a graph:




(3) They can be described by a recipe: say $a_n$ is the $n^{\text{th}}$ prime number for all $n \in \mathbb{N}$, or $a_1 = a_2 = 1$ and $a_{n+2} = a_n + a_{n+1}$ for all $n \in \mathbb{N}$, or say $a_1 = 2$ and

$$a_{n+1} = \frac{8}{a_n} + \frac{a_n}{2}$$

for all $n \in \mathbb{N}$.

**Definition 2.3.** Given $l \in \mathbb{R}$ and some sequence $(a_n)$, we say that $(a_n)$ *converges to the limit* $l \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$|a_n - l| < \varepsilon \quad \text{for all} \quad n \geqslant N.$$

We may write this as $\lim_{n\to\infty} a_n = l$ or $a_n \to l$ as $n \to \infty$, or simply, $a_n \to l$. We say that the sequence $(a_n)$ *converges* if there exists some $l \in \mathbb{R}$ such that $a_n \to l$.

Hence, if $a_n \to l$, then for whichever radius $\varepsilon > 0$ someone chooses, we can always find some $N \in \mathbb{N}$ such that $a_n \in (l - \varepsilon, l + \varepsilon)$ for all $n \geqslant N$.

The simplest possible example of a convergent sequence is the *constant sequence*. Thus, given $q \in \mathbb{R}$, define $a_n = q$ for every $n \in \mathbb{N}$. Then note that $a_n \to q$. Indeed, for any $\varepsilon > 0$, choosing $N = 1$, note that $|a_n - q| = 0 < \varepsilon$ for all $n \geqslant N = 1$.

**Lemma 2.4.** *We have $1/n \to 0$ as $n \to \infty$.*

*Proof.* We need to show that for all $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ (this $N$ may depend on $\varepsilon$) such that $|1/n - 0| < \varepsilon$ for all $n \geqslant N$. In particular, given some $\varepsilon > 0$, we need to have $1/\varepsilon < n$ for all $n \geqslant N$. Applying Archimedean principle, we see that there exists some $m \in \mathbb{N}$ such that $m > 1/\varepsilon$. Choosing $N = m$, we see that whenever $n \geqslant N$, we have

$$n \geqslant N = m > 1/\varepsilon$$

which is the desired conclusion. Thus, $1/n \to 0$ as $n \to \infty$. □

One can similarly show that for any $\alpha \in \mathbb{Q}$ with $\alpha > 0$, one has $1/n^\alpha \to 0$ as $n \to \infty$.

We will now show that if some sequence converges, then it converges to a unique limit.

**Lemma 2.5.** *Given some sequence $(a_n)$ and some real numbers $l_1, l_2$, if $a_n \to l_1$ and $a_n \to l_2$, then $l_1 = l_2$.*

*Proof.* Suppose not, and so, suppose $l_2, l_1$ are distinct. Thus, there must exist some $\varepsilon > 0$ such that the intervals $(l_1 - \varepsilon, l_1 + \varepsilon)$ and $(l_2 - \varepsilon, l_2 + \varepsilon)$ are disjoint; indeed choose $\varepsilon = |l_2 - l_1|/100$. But for this choice of $\varepsilon$, we know there exist $N_1, N_2 \in \mathbb{N}$ such that

$$a_n \in (l_1 - \varepsilon, l_1 + \varepsilon) \text{ for all } n \geqslant N_1, \quad \text{and} \quad a_n \in (l_2 - \varepsilon, l_2 + \varepsilon) \text{ for all } n \geqslant N_2.$$

Choose $n = \max\{N_1, N_2\} + 1$. Then $a_n \in (l_1 - \varepsilon, l_1 + \varepsilon) \cap (l_2 - \varepsilon, l_2 + \varepsilon)$, which contradicts the fact that these intervals were disjoint. □

The above proof can be rewritten as

$$|l_1 - l_2| \leqslant |l_1 - a_n| + |a_n - l_2| < 2\varepsilon = |l_1 - l_2|/50,$$

where the first inequality follows from triangle inequality.

*Lecture 8*

**Lemma 2.6.** *Let $x \in \mathbb{R}$ satisfy $|x| < 1$. Then $x^n \to 0$ as $n \to \infty$.*

*Proof.* Let $\varepsilon > 0$ be some real number. We need to find some $N \in \mathbb{N}$ such that $|x^n - 0| = |x|^n < \varepsilon$ for all $n \geqslant N$. Since $|x| = |-x|$ and since the result is trivial when $x = 0$, it suffices to assume that $0 < x < 1$.

Hence it suffices to find some $N \in \mathbb{N}$ such that $1/\varepsilon < (1/x)^n$ for all $n \geqslant N$. On the other hand, since $1/x > 1$, we may use Bernoulli's inequality to see that

$$\frac{1}{x^n} = \left(1 + \left(\frac{1}{x} - 1\right)\right)^n \geqslant 1 + n\left(\frac{1}{x} - 1\right) > n\left(\frac{1}{x} - 1\right). \tag{2.1}$$

Applying Archimedean principle , we can find some $N \in \mathbb{N}$ such that

$$N \geqslant \frac{\frac{1}{\varepsilon}}{\frac{1}{x} - 1}.$$

For any $n \geqslant N$, we may now use (2.1) to see that

$$\frac{1}{x^n} > n\left(\frac{1}{x} - 1\right) \geqslant N\left(\frac{1}{x} - 1\right) \geqslant \frac{1}{\varepsilon}.$$

This concludes the proof of Lemma 2.6. □

*Remark* 2.7. There are sequences that do not converge. For example, let $a_n = 2^n$ for all $n \in \mathbb{N}$, or let $a_n = 31$ when $n$ is even and $a_n = -1/2$ when $n$ is odd.

## 2.1. **Properties of limits.**

**Lemma 2.8** (Shift Rule)**.** *Let $(a_n)$ be a sequence, let $k \in \mathbb{N}$ be a fixed integer, and let $l \in \mathbb{R}$. Then $a_n \to l$ as $n \to \infty$ if and only if $a_{n+k} \to l$ as $n \to \infty$.*

*Proof.* Let $\varepsilon > 0$ be a real number. Let $b_n = a_{n+k}$ for all $n \in \mathbb{N}$.

Suppose $a_n \to l$. Then we know that there exists some $N \in \mathbb{N}$, possibly depending on $\varepsilon$, such that $|a_n - l| < \varepsilon$ for all $n \geqslant N$. Since $n + k \geqslant n$, this gives us that $|b_n - l| < \varepsilon$ for all $n \geqslant N$, and so, $b_n \to l$.

Now suppose that $b_n \to l$. This means that there exists some $N \in \mathbb{N}$, possibly depending on $\varepsilon$, such that $|b_n - l| = |a_{n+k} - l| < \varepsilon$ for all $n \geqslant N$. Rewriting this, we see that whenever $m \geqslant N + k$, we have $|a_m - l| < \varepsilon$. This means that $a_n \to l$. □

This means that $1/(n + 10) \to 0$ as $n \to \infty$ and $(1/6)^{n-5} \to 0$ as $n \to \infty$.

**Definition 2.9.** We say that a sequence $(a_n)$ is *bounded* if the set $\{a_n : n \in N\}$ is bounded, that is, there exists some $M > 0$ such that $|a_n| \leqslant M$ for all $n \in \mathbb{N}$.

**Lemma 2.10.** *Any convergent sequence is bounded.*

*Proof.* Let $(a_n)$ be a sequence and let $l$ be a real number such that $a_n \to l$. By definition of convergence, we know that there exists some $N \in \mathbb{N}$ such that whenever $n \geqslant N$, we have $|a_n - l| < 1$. By triangle inequality, this means that whenever $n \geqslant N$, we have $|a_n| \leqslant |a_n - l| + |l| < |l| + 1$. Setting

$$M = \max\{|a_1|, |a_2|, \ldots, |a_N|, |l| + 1\}$$

we see that $|a_n| \leqslant M$ for all $n \in \mathbb{N}$. □

*Remark* 2.11. Thus, if a sequence is not bounded, then it can not converge (consider $a_n = n + 5$ for all $n \in \mathbb{N}$). On the other hand, just because a sequence is bounded, does not mean that it must converge (consider $a_n = (-1)^n$ for all $n \in \mathbb{N}$).

**Lemma 2.12.** *Let $(a_n)$ be a sequence, let $l \in \mathbb{R}$ satisfy $a_n \to l$. Then $|a_n| \to |l|$.*

*Proof.* Let $\varepsilon > 0$ be a real number. Since $a_n \to l$, we know there exists some $N \in \mathbb{N}$ such that $|a_n - l| < \varepsilon$ whenever $n \geqslant N$. We can now use the reverse triangle inequality (1.5) to deduce that

$$||a_n| - |l|| \leqslant |a_n - l| < \varepsilon$$

whenever $n \geqslant N$. This implies that $|a_n| \to |l|$. □

*Lecture 9*

2.2. **Algebra of Limits.** There are some shortcuts that allow us to calculate limits swiftly without invoking Definition 2.3 every time.

**Lemma 2.13** (Algebra of Limits)**.** *Let $(a_n), (b_n)$ be sequences and $a, b$ be real numbers such that $a_n \to a$ and $b_n \to b$. Then*

*(1) $a_n + b_n \to a + b$*
*(2) $a_n b_n \to ab$*
*(3) for any $c \in \mathbb{R}$, we have $ca_n \to ca$*
*(4) if $b_n \neq 0$ for all $n \in \mathbb{N}$ and $b \neq 0$, then $1/b_n \to 1/b$.*
*(5) if $b_n \neq 0$ for all $n \in \mathbb{N}$ and $b \neq 0$, then $a_n/b_n \to a/b$.*

We will first give some examples where algebra of limits is useful.

**Example 2.14.** Let

$$a_n = \frac{n^5 + 16n}{15n^5 + 7n^2 + 1}$$

for all $n \in \mathbb{N}$. Note that this is the same as

$$a_n = \frac{1 + 16/n^4}{15 + 7/n^3 + 1/n^5},$$

and the denominator here is always positive. Since $16/n^4 \to 0$ and $7/n^3 \to 0$ and $1/n^5 \to 0$, by algebra of limits, we get that

$$1 + 16/n^4 \to 1 \quad \text{and} \quad 15 + 7/n^3 + 1/n^5 \to 15.$$

By using algebra of limits again, we see that

$$a_n \to 1/15.$$

**Example 2.15.** Let $a_1 = 2$ and let

$$a_{n+1} = \frac{8}{a_n} + \frac{a_n}{2}$$

for all $n \in \mathbb{N}$. Let's try to analyse some properties of this sequence first. We do see that $a_n > 0$ for all $n \in \mathbb{N}$. In fact, $a_{n+1} \geqslant 4$ for all $n \in \mathbb{N}$ since

$$a_{n+1} = \frac{8}{a_n} + \frac{a_n}{2} \geqslant 2\left(\frac{8}{a_n}\frac{a_n}{2}\right)^{1/2} = 4,$$

where we have used the AM-GM inequality: $a+b \geqslant 2(ab)^{1/2}$ whenever $a, b \geqslant 0$. This follows from the fact that $(a^{1/2} - b^{1/2})^2 \geqslant 0$ for all $a, b \geqslant 0$.

IF this sequence did converge, say to some limit $l$, then we see that $l > 0$ (since $a_n \geqslant 4$ for all $n \geqslant 2$; see also Lemma 2.18). This means that $8/a_n \to 8/l$ and $a_n/2 \to l/2$ via algebra of limits. Applying algebra of limits again, we get that

$$l = 8/l + l/2, \quad \text{that is,} \quad l^2 = 16.$$

Since we know $l > 0$, we would have that $l = 4$.

Now, let's try to prove that $a_n \to 4$. Let $n \geqslant 2$ be some integer. Note that

$$a_{n+1} - 4 = \frac{a_n}{2} + \frac{8}{a_n} - 4 = \frac{a_n^2 - 8a_n + 16}{2a_n} = \frac{(a_n - 4)^2}{2a_n} = (a_n - 4)\left(\frac{1}{2} - \frac{2}{a_n}\right).$$

Since $a_n \geqslant 4$, we thus have $0 < 2/a_n \leqslant 1/2$, and so,

$$|a_{n+1} - 4| \leqslant \frac{|a_n - 4|}{2}.$$

Iterating this, we get that for all $n \geqslant 2$, one has

$$|a_n - 4| \leqslant \frac{|a_{n-1} - 4|}{2} \leqslant \frac{|a_{n-2} - 4|}{2^2} \leqslant \ldots \leqslant \frac{|a_2 - 4|}{2^{n-2}} = \frac{1}{2^{n-2}}.$$

Finally, given $\varepsilon > 0$, choosing $N \in \mathbb{N}$ such that $N > 1/\varepsilon + 1$ (such an $N$ exists via the Archimedean principle), we see that whenever $n \geqslant N$, one has

$$2^{n-2} \geqslant 1 + (n-2) \geqslant N - 1 > 1/\varepsilon,$$

where the first step follows from Bernoulli's inequality. Noting the preceding inequality, this gives us $|a_n - 4| \leqslant 1/2^{n-2} < \varepsilon$ for all $n \geqslant N$. We thus conclude that $a_n \to 4$.

*Remark* 2.16. We will briefly comment on the AM–GM (Arithmetic Mean–Geometric Mean) inequality. Given $n \in \mathbb{N}$, and some non-negative real numbers $a_1, \dots, a_n$, one has

$$\frac{a_1 + \cdots + a_n}{n} \geqslant (a_1 \dots a_n)^{1/n}. \tag{2.2}$$

*Proof (Non-examinable).* We will prove this for all $n \in \mathbb{N}$. Note that if $a_i = 0$ for any $1 \leqslant i \leqslant n$, then the right hand side is 0 while the left hand side is non-negative, and so we are done trivially. Hence, we may assume that $a_i > 0$ for all $1 \leqslant i \leqslant n$.

Inequality (2.2) is trivial when $n = 1$. When $n = 2^k$ for some $k \in \mathbb{N}$, we can prove this via induction on $k$. Indeed, the base case is trivial:

$$\frac{a_1 + a_2}{2} \geqslant (a_1 a_2)^{1/2} \quad \Leftrightarrow \quad a_1 + a_2 - 2(a_1 a_2)^{1/2} \geqslant 0 \quad \Leftrightarrow \quad (a_1^{1/2} - a_2^{1/2})^2 \geqslant 0.$$

Given $k \in \mathbb{N}$, suppose this holds true when $n = 2^k$. Thus given non-negative real numbers $a_1, \dots, a_{2^{k+1}}$, one has

$$\begin{aligned} \frac{a_1 + \cdots + a_{2^{k+1}}}{2^{k+1}} &= \frac{1}{2}\left(\frac{a_1 + \cdots + a_{2^k}}{2^k} + \frac{a_{2^k+1} + \cdots + a_{2^{k+1}}}{2^k}\right) \\ &\geqslant \frac{1}{2}\left((a_1 \dots a_{2^k})^{1/2^k} + (a_{2^k+1} \dots a_{2^{k+1}})^{1/2^k}\right) \\ &\geqslant (a_1 \dots a_{2^k} a_{2^k+1} \dots a_{2^{k+1}})^{1/2^{k+1}}, \end{aligned}$$

where the second step follows from the inductive hypothesis when $n = 2^k$ and the third step follows from the case when $n = 2$.

Now suppose $2^k < n < 2^{k+1}$ for some $k \in \mathbb{N}$. We can use something called 'forward–backward–induction'. Let

$$\alpha = (a_1 + \cdots + a_n)/n \geqslant 0.$$

Since we know the AM–GM inequality holds when we have $2^{k+1}$ many elements, we see that

$$a_1 + \cdots + a_n + (2^{k+1} - n)\alpha \geqslant 2^{k+1}(a_1 \dots a_n \alpha^{2^{k+1}-n})^{1/2^{k+1}}.$$

On the other hand, the left hand side equals

$$a_1 + \cdots + a_n + (2^{k+1} - n)\alpha = n\alpha + (2^{k+1} - n)\alpha = 2^{k+1}\alpha.$$

Combining this with the above inequality, we find that

$$2^{k+1}\alpha \geqslant 2^{k+1}(a_1 \dots a_n \alpha^{2^{k+1}-n})^{1/2^{k+1}} \quad \Leftrightarrow \quad \alpha^{2^{k+1}} \geqslant a_1 \dots a_n \alpha^{2^{k+1}-n} \quad \Leftrightarrow \quad \alpha^n \geqslant a_1 \dots a_n.$$

The last inequality above is the claimed estimate, and so, we are done. □

We will now prove Lemma 2.13.

*Proof of Lemma 2.13.* Let $\varepsilon > 0$ be a real number.

(1) We will first prove (1). Since $a_n \to a$ and $b_n \to b$, there exist $N_1, N_2 \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon/2 \text{ for all } n \geqslant N_1, \quad \text{and} \quad |b_n - b| < \varepsilon/2 \text{ for all } n \geqslant N_2.$$

Let $N = \max\{N_1, N_2\}$. Then whenever $n \geqslant N$, we may apply triangle inequality to deduce that

$$|a_n + b_n - a - b| \leqslant |a_n - a| + |b_n - b| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

This proves that $a_n + b_n \to a + b$.

(2) We will now prove (2). Since $(a_n), (b_n)$ are convergent sequences, Lemma 2.10 implies that they are bounded, and so, there exists some $M > 0$ such that

$$a, b, |a_n|, |b_n| \leqslant M \text{ for all } n \in \mathbb{N}.$$

Now, since $a_n \to a$ and $b_n \to b$, we see that there exist $N_3, N_4 \in \mathbb{N}$ such that

$$|a_n - a| < \varepsilon/(2M) \text{ for all } n \geqslant N_3, \quad \text{and} \quad |b_n - b| < \varepsilon/(2M) \text{ for all } n \geqslant N_4.$$

Thus implies that whenever $n \geqslant N = \max\{N_3, N_4\}$, we have

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - ab_n + ab_n - ab| \leqslant |(a_n - a)b_n| + |a(b_n - b)| \\ &\leqslant |a_n - a||b_n| + |a||b_n - b| < (\varepsilon/2M)M + M(\varepsilon/2M) \\ &= \varepsilon/2 + \varepsilon/2 = \varepsilon. \end{aligned}$$

Thus $a_n b_n \to ab$.

(3) We see that(3) follows from (2) by setting $b_n = c$ for all $n \in \mathbb{N}$.

(4) Thus, we now set to prove (4). We will firstly need to show that the elements $b_n, b$ are bounded away from 0 whenever $n$ is sufficiently large. In order to see this, use the facts that $b_n \to b$ and $b \neq 0$ to deduce that there exists some $N_5 \in \mathbb{N}$ such that whenever $n \geqslant N_5$, one has $|b_n - b| < |b|/2$. This combines with triangle inequality to give us that

$$|b_n| > |b| - |b_n - b| > |b|/2 > 0,$$

whenever $n \geqslant N_5$. Now since $b_n \to b$, we see that there exists some $N_6 \in \mathbb{N}$ such that

$$|b_n - b| < \frac{\varepsilon |b|^2}{2}$$

whenever $n \geqslant N_6$. Thus, setting $N = \max\{N_5, N_6\}$, we see that whenever $n \geqslant N$, we have

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| = \left|\frac{b_n - b}{bb_n}\right| < \frac{2|b_n - b|}{|b|^2} < \varepsilon.$$

Thus, we have that $1/b_n \to 1/b$.

(5) Finally, in the context of (5), we may combine (2) and (4) to deduce that $a_n/b_n \to a/b$.

□

*Remark* 2.17. Let $(a_n), (b_n), (c_n)$ be sequences, let $a, b, c \in \mathbb{R}$ satisfy $a_n \to a$ and $b_n \to b$ and $c_n \to c$. We can use algebra of limits twice to show that

$$\lim_{n\to\infty}(a_n + b_n + c_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty}(b_n + c_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n + \lim_{n\to\infty} c_n = a + b + c.$$

Similarly, we can show that $\lim_{n\to\infty} a_n b_n c_n = abc$.

*Lecture 11*

2.3. **Limits and inequalities.** Strict inequalities are not preserved by limits, indeed, we have $1 - 1/n < 1$ for all $n \in \mathbb{N}$, but $\lim_{n\to\infty}(1 - 1/n) = 1$.

**Lemma 2.18.** *Let $(a_n), (b_n)$ be sequences, let $N \in \mathbb{N}$ be such that $a_n \leqslant b_n$ for all $n \geqslant N$. Let $a_n \to a$ and $b_n \to b$ for some $a, b \in \mathbb{R}$. Then $a \leqslant b$.*

*Proof.* We can prove this by contradiction, and so, suppose $b < a$. Let $\varepsilon = (a - b)/100 > 0$. For this choice of $\varepsilon$, there exist $N_1, N_2 \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ whenever $n \geqslant N_1$ and $|b_n - b| < \varepsilon$ whenever $n \geqslant N_2$. This means that whenever $n \geqslant \max\{N, N_1, N_2\}$, we have

$$a_n > a - \varepsilon > b + \varepsilon > b_n \geqslant a_n,$$

delivering the desired contradiction. □

*Remark* 2.19. This immediately implies that if a sequence $(a_n)$, some integer $N \geqslant 1$, some real numbers $l_1 < l_2$ satisfy $l_1 \leqslant a_n \leqslant l_2$ for all $n \geqslant N$, and if $a_n \to l$ for some $l \in \mathbb{R}$, then $l_1 \leqslant l \leqslant l_2$.

**Lemma 2.20** (Sandwich rule)**.** *Let $(a_n), (b_n), (c_n)$ be sequences, let $N \in \mathbb{N}$ be such that $a_n \leqslant b_n \leqslant c_n$ for all $n \geqslant N$. Suppose $a_n \to l$ and $c_n \to l$ for some $l \in \mathbb{R}$. Then $b_n \to l$.*

*Proof.* Let $\varepsilon > 0$ be a real number. Since $a_n \to l$ and $c_n \to l$, we may find some $N' \in \mathbb{N}$ such that $|a_n - l| < \varepsilon$ and $|c_n - l| < \varepsilon$ whenever $n \geqslant N'$. Thus, whenever $n \geqslant \max\{N', N\}$, we

get that

$$l - \varepsilon < a_n \leqslant b_n \leqslant c_n < l + \varepsilon,$$

that is, $|b_n - l| < \varepsilon$. This implies that $b_n \to l$. □

This has many applications in calculating limits.

**Lemma 2.21.** *Let $x > 0$. Then $x^{1/n} \to 1$ as $n \to \infty$.*

*Proof.* This holds true trivially when $x = 1$. We first consider the case when $x > 1$. Since $x > 1$, Lemma 0.5 implies that $x^{1/n} > 1$. Writing $h_n = x^{1/n} - 1$ for all $n \in \mathbb{N}$, we see that $h_n > 0$. Moreover, we may apply Bernoulli's inequality to get that

$$x = (1 + h_n)^n \geqslant 1 + nh_n > nh_n.$$

Simplifying, we see that $0 < h_n < x/n$ for all $n \in \mathbb{N}$. Since $x > 1$ is fixed, $x/n \to 0$. Sandwich rule now implies that $h_n \to 0$. We can now apply algebra of limits to deduce that $x^{1/n} \to 1$.

Now suppose $0 < x < 1$. Since

$$x^{1/n} = \frac{1}{(1/x)^{1/n}},$$

and $1/x > 1$, we can apply algebra of limits and the fact that $(1/x)^{1/n} \to 1$ as $n \to \infty$ to deduce that $x^{1/n} \to 1$ as $n \to \infty$. □

**Lemma 2.22.** *We have $n^{1/n} \to 1$ as $n \to \infty$.*

*Proof.* It suffices to show that $n^{1/(2n)} \to 1$ as $n \to \infty$ because then the desired estimate follows from algebra of limits and the fact that $n^{1/n} = n^{1/(2n)} \cdot n^{1/(2n)}$. In order to show the former, as before, let $h_n = n^{1/(2n)} - 1$. For all $n \in \mathbb{N}$, we see that $h_n \geqslant 0$. Moreover, applying Bernoulli's inequality, we see that

$$n^{1/2} = (1 + h_n)^n \geqslant 1 + nh_n > nh_n,$$

that is, $h_n < 1/n^{1/2}$. Since $1/n^{1/2} \to 0$, we may apply Sandwich rule to deduce that $h_n \to 0$ as $n \to \infty$. Algebra of limits now tells us that $n^{1/(2n)} \to 1$ as $n \to \infty$. □

**Example* 2.23.** We will now provide an alternative proof of the fact that there is no $x \in \mathbb{Q}$ such that $x^2 = 2$.

*Proof (non-examinable).* We prove this via contradiction, and so, suppose $x = p/q$ satisfies $x^2 = 2$, with $p, q$ being coprime integers. By replacing $x$ with $-x$ if necessary and noting that $0^2 = 0$, we may assume that $p, q \in \mathbb{N}$. Now note that since $1^2 < 2 < 2^2$, we must have

$1 < x < 2$. Thus, for any $n \in \mathbb{N}$, we have

$$(x-1)^n = \sum_{j=0}^{n} \binom{n}{j} x^j (-1)^{n-j} = a_n + b_n x = \frac{a_n q + b_n p}{q} \tag{2.3}$$

for some $a_n, b_n \in \mathbb{Z}$. Here we have used the fact that $x^2 = 2$, and so, for any $j \in \mathbb{N}$, either $x^j$ is an integer, or $x^j$ is some integer multiple of $x$. Since $x > 1$ and $q \geqslant 1$, we deduce that $a_n q + b_n p \in \mathbb{N}$, and so, the right hand side in (2.3) is always at least $1/q$. Since limits preserve inequalities, we get that

$$\lim_{n\to\infty} (x-1)^n \geqslant 1/q.$$

On the other hand, since $0 < x - 1 < 1$, Lemma 2.6 implies that $\lim_{n\to\infty}(x-1)^n = 0$, thus delivering the required contradiction. □

2.4. **Powers and factorials.**

**Definition 2.24.** Given a sequence $(a_n)$, we say that $a_n \to \infty$ if for every $R > 0$, there exists some $N \in \mathbb{N}$ such that $a_n > R$ for all $n \geqslant N$. Similarly, we say that $a_n \to -\infty$ if for every $R > 0$, there exists some $N \in \mathbb{N}$ such that $a_n < -R$ for all $n \geqslant N$.

*Remark* 2.25. A sequence $(a_n)$ may be unbounded and still may not satisfy $a_n \to \infty$ or $a_n \to -\infty$. Indeed, consider the sequence $a_n$, where $a_n = n$ when $n = 3k-2$ for some $k \in \mathbb{N}$, and $a_n = -n$ when $n = 3k-1$ for some $k \in \mathbb{N}$, and $a_n = 0$ when $n = 3k$ for some $k \in \mathbb{N}$.

**Example 2.26.** Here are some simple examples.

(1) $n \to \infty$ as $n \to \infty$. Indeed, for any $R > 0$, choose $N = \lfloor R \rfloor + 1$, and note that for any $n \geqslant N$, we have $n \geqslant N > R$.

(2) $n^{1/2} \to \infty$ as $n \to \infty$. Indeed, for any $R > 0$, choose $N = \lfloor R^2 + 100 \rfloor$.

(3) For any fixed $x > 1$, we have

$$x^n \to \infty \quad \text{as} \quad n \to \infty. \tag{2.4}$$

We can apply Bernoulli's inequality to deduce that $x^n \geqslant 1 + n(x-1) > n(x-1)$. Given $R > 0$, we can choose $N = \lfloor R/(x-1) \rfloor + 10$ and see that

$$x^n \geqslant n(x-1) \geqslant N(x-1) > R$$

holds for any $n \geqslant N$.

**Lemma 2.27.** *Suppose $(a_n)$ satisfies $a_n \neq 0$ for all $n \in \mathbb{N}$ and $a_n \to \infty$ as $n \to \infty$. Then $1/a_n \to 0$ as $n \to \infty$.*

*Proof.* Let $\varepsilon > 0$ be a real number. It suffices to show that there exists some $N \in \mathbb{N}$ such that $|1/a_n| < \varepsilon$ whenever $n \geqslant N$. Now, since $a_n \to \infty$, there must exist some $N' \in \mathbb{N}$ such that $a_n > 1/\varepsilon$ for all $n \geqslant N'$. Choose $N = N'$ and note that whenever $n \geqslant N$, we have $|1/a_n| = 1/a_n < \varepsilon$. □

**Lemma 2.28** (Comparison test for sequences)**.** *If $(a_n), (b_n)$ are sequences such that $b_n \to \infty$ as $n \to \infty$, and $N \in \mathbb{N}$ satisfies $a_n \geqslant b_n$ for all $n \geqslant N$, then in fact $a_n \to \infty$ as $n \to \infty$.*

*Proof.* Given $R > 0$, there exists $N' \in \mathbb{N}$ such that $b_n \geqslant R$ for all $n \geqslant N'$. Choose $N'' = \max\{N, N'\}$ and note that whenever $n \geqslant N''$, we have $a_n \geqslant R$. □

While this is simple to prove, it is a powerful tool that allows us to swiftly deduce limits in many cases. For instance, this allows us to prove the so-called 'Ratio test'.

**Lemma 2.29** (Ratio test for limits)**.** *Let $(a_n)$ be a sequence such that $a_n > 0$ for all $n \in \mathbb{N}$. Moreover, suppose that $a_{n+1}/a_n \to r$ as $n \to \infty$ for some $r \in \mathbb{R}$. If $r < 1$, then $a_n \to 0$, and if $r > 1$, then $a_n \to \infty$.*

*Proof.* We first consider the case when $r < 1$. Since $a_{n+1}/a_n \to r$ as $n \to \infty$, we see that there exists some $N \in \mathbb{N}$ such that for all $n \geqslant N$, one has

$$\frac{a_{n+1}}{a_n} \in \left(r - \frac{1-r}{2}, r + \frac{1-r}{2}\right).$$

In particular, we have $a_{n+1} \leqslant (r+1)a_n/2$ for all $n \geqslant N$. Writing $q = (r+1)/2 < 1$, we see that

$$0 < a_n \leqslant q a_{n-1} \leqslant q^2 a_{n-2} \leqslant \ldots \leqslant q^{n-N} a_N$$

for all $n \geqslant N$. Note that $N$ is some fixed natural number and $q < 1$, and so, Lemma 2.6 and algebra of limits imply that $q^{n-N} a_N \to 0$ as $n \to \infty$. We may now use Sandwich rule to deduce that $a_n \to 0$ as $n \to \infty$.

We now deal with the case when $r > 1$. As before, since $a_{n+1}/a_n \to r$, we see that there must exist some $N' \in \mathbb{N}$ such that

$$\frac{a_{n+1}}{a_n} \in \left(r - \frac{r-1}{2}, r + \frac{r-1}{2}\right)$$

for all $n \geqslant N'$. In particular, writing $p = (r+1)/2 > 1$, we see that $a_{n+1} \geqslant p a_n$ for all $n \geqslant N'$. Iterating this as before, we get that

$$a_n \geqslant p^{n-N'} a_{N'}$$

for all $n \geqslant N'$. Since $N'$ is some fixed natural number, $p > 1$ and $a_{N'} > 0$, we can apply algebra of limits and (2.4) to deduce that $p^{n-N'}a_{N'} \to \infty$ as $n \to \infty$. We may now use Lemma 2.28 to deduce that $a_n \to \infty$ as $n \to \infty$. □

We now prove the following standard limits.

**Proposition 2.30.** *Let $x \in \mathbb{R}$ and let $\alpha \in \mathbb{Q}$ and let $k \in \mathbb{N}$. The following hold true as $n \to \infty$.*

*(1) If $\alpha > 0$, then $n^\alpha \to \infty$. If $\alpha < 0$, then $n^\alpha \to 0$.*
*(2) If $|x| > 1$, then $|x|^n \to \infty$. If $|x| < 1$, then $x^n \to 0$.*
*(3) If $x > 1$, then $x^n/n^k \to \infty$. If $0 \leqslant x \leqslant 1$, then $x^n/n^k \to 0$.*
*(4) We have $x^n/n! \to 0$.*
*(5) $n!/n^n \to 0$.*

In particular, given $x > 1$ and $k \in \mathbb{N}$, there is a hierarchy amongst the functions $n^k$, $x^n$, $n!$ and $n^n$, wherein, each following function grows much faster than the preceding ones.

*Proof of Proposition 2.30.* We divide the proof as follows.

(1) We first prove (1). If $\alpha > 0$, we may write $\alpha = p/q$ for some fixed $p, q \in \mathbb{N}$. Note that $p$ is a fixed natural number and $n^{p/q} = n^{1/q} \dots n^{1/q}$ where we have $p$ many terms on the right hand side. Thus, we may use the product rule in Algebra of limits (see Remark 2.17) to see that it suffices to prove that $n^{1/q} \to \infty$ as $n \to \infty$. In order to show the latter, note that it suffices to prove that for any $R > 0$, there exists some $N \in \mathbb{N}$ such that $N > R^q$. One can do this via the Archimedean principle.

If $\alpha < 0$, we may use the fact that $n^{-\alpha} \to \infty$ and Lemma 2.27 to deduce that $n^\alpha = 1/(n^{-\alpha}) \to 0$.

(2) We now consider (2). We prove the first part, and so, let $a_n = |x|^n$ for all $n \in \mathbb{N}$. Note that we can apply the ratio test here since $a_n > 0$ for all $n \in \mathbb{N}$. In this case, we see that $a_{n+1}/a_n = |x|$, and so $a_{n+1}/a_n \to |x|$ where $|x| > 1$. Thus by the ratio test, $a_n \to \infty$ as $n \to \infty$.

When $|x| < 1$, we note that $|x|^n = 1/(1/|x|)^n$, and since $1/|x| > 1$, we see that $(1/|x|)^n \to \infty$. Lemma 2.27 now implies that $|x|^n \to 0$.

(3) We now turn to (3). Note that if $x = 0$, then $x^n/n^k = 0$ for all $n \in \mathbb{N}$, in which case $x^n/n^k \to 0$ trivially. Thus suppose that $x > 0$. Let $a_n = x^n/n^k$ for all $n \in \mathbb{N}$. Since

$a_n > 0$ for all $n \in \mathbb{N}$, we can now apply the ratio test. In particular, note that

$$\frac{a_{n+1}}{a_n} = \frac{xn^k}{(n+1)^k} = \frac{x}{(1+1/n)^k}.$$

We may now apply algebra of limits and the fact that $k \in \mathbb{N}$ is fixed to deduce that

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{x}{(1+1/n)^k} = \frac{x}{(1+\lim_{n\to\infty} 1/n)^k} = x.$$

Thus by the ratio test, if $x > 1$, then $x^n/n^k \to \infty$, and if $x < 1$, then $x^n/n^k \to 0$. Finally, if $x = 1$, then $x^n/n^k = 1/n^k \to 0$ as $n \to \infty$.

(4) We now proceed to prove (4). This is trivial when $x = 0$, and so, we assume that $x \neq 0$. Note that $-|x|^n \leqslant x^n \leqslant |x|^n$ for any $x \in \mathbb{R}$, and so, it suffices to prove that $|x|^n/n! \to 0$ as $n \to \infty$ and then apply the Sandwich rule to deduce the desired conclusion. Thus, let $a_n = |x|^n/n!$ for every $n \in \mathbb{N}$ and note that $a_n > 0$ for all $n \in \mathbb{N}$. Moreover, we have

$$\frac{a_{n+1}}{a_n} = \frac{|x|}{n+1} \to 0$$

as $n \to \infty$. Therefore, the ratio test implies that $a_n \to 0$ as $n \to \infty$.

(5) We will finally prove (5). Note that for every $n \geqslant 3$, we have

$$0 < \frac{n!}{n^n} = \frac{1}{n} \cdot \frac{2}{n} \cdots \frac{n-1}{n} \cdot \frac{n}{n} < \frac{1}{n}.$$

Since $1/n \to 0$, we may apply the Sandwich rule to deduce that $n!/n^n \to 0$ as $n \to \infty$.

□

*Lecture 13*

2.5. **Monotonic sequences.** In most of the above cases, we have proven convergence for sequences that had some form of explicit description. It turns out that often this is not required, in fact, we can sometimes prove that a sequence converges without even knowing the limit.

**Definition 2.31.** A sequence $(a_n)$ is *increasing* if $a_n \leqslant a_{n+1}$ for all $n \in \mathbb{N}$. A sequence is *strictly increasing* if $a_n < a_{n+1}$ for all $n \in \mathbb{N}$.

Similarly, a sequence $(a_n)$ is *decreasing* if $a_n \geqslant a_{n+1}$ for all $n \in \mathbb{N}$, and $(a_n)$ is *strictly decreasing* if $a_n > a_{n+1}$ for all $n \in \mathbb{N}$.

If a sequence is increasing or decreasing, we call it *monotonic*. Is a sequence is strictly increasing or strictly decreasing, we call it *strictly monotonic*.

A sequence $(a_n)$ is *bounded above* if the set $\{a_n : n \in \mathbb{N}\}$ is bounded above, that is, if there exists some $M \in \mathbb{R}$ such that $a_n \leqslant M$ for all $n \in \mathbb{N}$. Similarly, a sequence $(a_n)$ is *bounded below* if the set $\{a_n : n \in \mathbb{N}\}$ is bounded below. A sequence is *bounded* if it is both bounded above and bounded below.

**Theorem 2.32.** *Suppose $(a_n)$ is an increasing sequence. If $(a_n)$ is bounded above, then it converges. If $(a_n)$ is not bounded above, then $a_n \to \infty$.*

*Proof.* We begin by proving the first part. Let $S = \{a_n : n \in \mathbb{N}\}$. Since $S$ is non-empty and bounded above, by the least upper bound axiom, it has a least upper bound. Let $l = \sup S$. By definition, for every $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that $l - \varepsilon < a_N \leqslant l$. Since $(a_n)$ is increasing, we see that for all $n \geqslant N$, one has $l - \varepsilon < a_N \leqslant a_n$. Moreover, since $l$ is an upper bound for $S$, we see that $a_n \leqslant l$ for all $n \in N$. Thus, we have just shown that for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geqslant N$, one has $l - \varepsilon < a_n \leqslant l$. This means that $a_n \to l$.

We now prove the second part. Writing $S = \{a_n : n \in \mathbb{N}\}$, since $S$ is not bounded above, we see that for any $R \in \mathbb{R}$, there must exist some $N \in \mathbb{N}$ such that $a_N > R$. On the other hand, since $(a_n)$ is increasing, we have that for any $n \geqslant N$, the inequality $a_n \geqslant a_N > R$ holds. Thus, $a_n \to \infty$. □

*Remark* 2.33. The first part of (2.32) is in fact equivalent to the least upper bound axiom.

*Remark* 2.34. Note that the sequence $(a_n)$ is increasing if and only if the sequence $(-a_n)$ is decreasing.

The above remark may be combined with Theorem 2.32 to deduce the following.

**Corollary 2.35.** *Let $(a_n)$ be a decreasing sequence. If $(a_n)$ is bounded below, then it converges, and if $(a_n)$ is not bounded below, then $a_n \to -\infty$.*

**Example 2.36.** Recall the sequence $(a_n)$ in Example 2.15, that is, let $a_1 = 2$ and let

$$a_{n+1} = \frac{8}{a_n} + \frac{a_n}{2}$$

for all $n \in \mathbb{N}$. We will now show a shorter proof that $(a_n)$ converges.

Let $b_n = a_{n+1}$ for all $n \in \mathbb{N}$. In Example 2.15, we proved that $b_n \geqslant 4$ for all $n \in \mathbb{N}$ by the AM–GM inequality. Moreover, note that

$$b_{n+1} - b_n = \frac{8}{b_n} - \frac{b_n}{2} = \frac{16 - b_n^2}{2b_n} < 0$$

since $b_n \geqslant 4$ for all $n \in N$. Thus, the sequence $b_n$ is decreasing and bounded below, and so, Corollary 2.35 implies that $(b_n)$ converges. Using algebra of limits, we can show that $b_n \to 4$. Applying the shift rule, we get that $a_n \to 4$.

**Example 2.37.** Let $a_1 = 1$ and let $a_{n+1} = (a_n + 2)^{1/2}$ for all $n \in \mathbb{N}$. One can show by induction that $a_n > 0$ for all $n \in \mathbb{N}$, which, in fact, would further imply that

$$a_{n+1} = (a_n + 2)^{1/2} > 2^{1/2}$$

for all $n \in \mathbb{N}$.

Now, IF the sequence $(a_n)$ converged, say, $a_n \to l$ for some $l \in \mathbb{R}$, then by algebra of limits, we would get that $l^2 - l - 2 = 0$, that is, either $l = 2$ or $l = -1$. Since $a_n > 0$, we would get that $l = 2$.

With this intuition, let's check whether $(a_n)$ is increasing or decreasing. Thus, note that

$$a_{n+1}^2 - a_n^2 = a_n + 2 - a_n^2 = -(a_n - 2)(a_n + 1).$$

Note that $a_1 = 1 < 2$. If $a_n \leqslant 2$ for all $n \in \mathbb{N}$, then we would get that the right hand side is always non-negative, that is, $a_{n+1}^2 \geqslant a_n^2$ for all $n \in \mathbb{N}$. Since $a_n > 0$ for all $n \in \mathbb{N}$, this would mean that $a_{n+1} \geqslant a_n$ for all $n \in \mathbb{N}$.

Thus, it now just suffices to prove that $a_n \leqslant 2$ for all $n \in \mathbb{N}$. We can prove this via induction, indeed the case when $n = 1$ is trivial. Now suppose $a_k \leqslant 2$ for some $k \in \mathbb{N}$. Assuming this, we get that

$$a_{k+1} = (a_k + 2)^{1/2} \leqslant 4^{1/2} = 2,$$

which concludes the inductive step.

Hence, we have shown that $a_n \leqslant 2$ for all $n \in \mathbb{N}$, which in turn gives us $(a_n)$ is increasing. By Theorem 2.32, we get that $(a_n)$ converges, and finally, this would mean that $a_n \to 2$.

**Example 2.38.** Recalling the set described in item (5) of Example 1.5, we define the sequence $a_n = (1 + 1/n)^n$ for all $n \in \mathbb{N}$. There, we showed that $2 \leqslant a_n \leqslant 3$ for all $n \in \mathbb{N}$. Moreover, note that

$$a_n^{-1} = \frac{n^n}{(n+1)^n} = \left(1 - \frac{1}{n+1}\right)^n.$$

This is useful because

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n+1}\right)^{n+1}\left(1 - \frac{1}{n+1}\right)^n = \left(1 + \frac{1}{n+1}\right)\left(1 - \frac{1}{(n+1)^2}\right)^n.$$

for all $n \in \mathbb{N}$. Applying Bernoulli's inequality, we get that

$$\frac{a_{n+1}}{a_n} \geqslant \left(1 + \frac{1}{n+1}\right)\left(1 - \frac{n}{(n+1)^2}\right) = 1 + \frac{1}{(n+1)^3} > 1$$

for all $n \in \mathbb{N}$. Thus the sequence $(a_n)$ is bounded and increasing, and so, by Theorem 2.32, $(a_n)$ converges.

*Lecture 14*

2.6. **Cauchy sequences and the Bolzano–Weierstrass Theorem.**

**Definition 2.39.** A sequence $(a_n)$ is *Cauchy* if for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that whenever $n, m \geqslant N$, then

$$|a_n - a_m| < \varepsilon.$$

**Lemma 2.40.** *Let $(a_n)$ be some sequence and let $l \in \mathbb{R}$ satisfy $a_n \to l$ as $n \to \infty$. Then $(a_n)$ is Cauchy.*

*Proof.* Since $a_n \to l$, we may find some $N \in \mathbb{N}$ such that for all $n \geqslant N$, one has $|a_n - l| < \varepsilon/2$. Note that for any $n, m \geqslant N$, one can now apply the triangle inequality to deduce that

$$|a_n - a_m| \leqslant |a_n - l| + |l - a_m| < \varepsilon/2 + \varepsilon/2 = \varepsilon. \qquad \square$$

**Lemma 2.41.** *Cauchy sequences are bounded.*

*Proof.* Let $(a_n)$ be a Cauchy sequence. Thus, there exists some $N \in \mathbb{N}$ such that for all $n, m \geqslant N$, one has $|a_n - a_m| < 1/2$. In particular, this means that for all $n \geqslant N$, one has

$$|a_n| \leqslant |a_N| + |a_n - a_N| < |a_N| + 1/2.$$

Writing

$$M = \max\{|a_1|, |a_2|, \ldots, |a_{N-1}|, |a_N| + 1\},$$

we see that $|a_n| \leqslant M$ for all $n \in \mathbb{N}$. $\square$

This notion is quite helpful. For instance, this gives us a swift way to prove that the sequence $(a_n)$ satisfying $a_n = (-1)^n$ does not converge. Indeed, for any $N \in \mathbb{N}$ and any $n \geqslant N$, one has $|a_n - a_{n+1}| = 2$, whence, $(a_n)$ is not Cauchy. Lemma 2.40 now tells us that $(a_n)$ can not be convergent.

On the other hand, the above sequence can be interpreted as a union of two convergent sequences. Indeed, let $b_n = a_{2n-1}$ for all $n \in \mathbb{N}$ and let $c_n = a_{2n}$ for all $n \in \mathbb{N}$. Then $b_n \to -1$ and $c_n \to 1$. Thus, even if a sequence does not converge, it may contain a convergent "subsequence".

**Definition 2.42.** Let $(a_n)_{n=1}^{\infty}$ be a sequence. Let $(n_j)_{j=1}^{\infty}$ be a sequence such that $n_j \in \mathbb{N}$ for all $j \in \mathbb{N}$ and $n_{j+1} > n_j$ for all $j \in \mathbb{N}$. This gives us a *subsequence* $(a_{n_j})_{j=1}^{\infty}$ of $(a_n)_{n=1}^{\infty}$. The $j^{\text{th}}$ term of this subsequence is precisely $a_{n_j}$.

*Remark* 2.43. Let $(n_j)_{j=1}^{\infty}$ be a sequence such that $n_j \in \mathbb{N}$ for all $j \in \mathbb{N}$ and $n_{j+1} > n_j$ for all $j \in \mathbb{N}$. Then $(n_j)_{j=1}^{\infty}$ is itself a subsequence of the sequence $(n)_{n=1}^{\infty}$ of natural numbers.

Moreover, one can show that $n_j \to \infty$ as $j \to \infty$. In fact, we can prove the stronger statement that $n_j \geqslant j$ for all $j \in \mathbb{N}$. In order to do the latter, we can apply induction. For the base case, note that since $n_1 \in \mathbb{N}$, we must have $n_1 \geqslant 1$. Proceeding to the inductive step, suppose that $n_j \geqslant j$ for some $j \in \mathbb{N}$. Then since $n_{j+1} > n_j \geqslant j$ and since $n_{j+1} \in \mathbb{N}$, we must have that $n_{j+1} \geqslant j + 1$. This concludes the inductive step.

**Example 2.44.** The sequence $0, 1, 0, 2, 0, 3, 0, 4, 0, \ldots$ has the following subsequences:

$$0, 0, 0, 0, 0, \ldots \quad \text{and} \quad 1, 2, 3, 4, 5, \ldots \quad \text{and} \quad 1, 0, 4, 0, 7, 0, 10, \ldots \quad \text{and} \quad 3, 9, 27, 81, 243, \ldots$$

**Lemma 2.45.** *Suppose $(a_n)_{n=1}^{\infty}$ is a sequence and $l \in \mathbb{R}$ satisfies $a_n \to l$. Then every subsequence $(a_{n_j})_{j=1}^{\infty}$ of $(a_n)_{n=1}^{\infty}$ also satisfies $a_{n_j} \to l$ as $j \to \infty$.*

*Proof.* Let $\varepsilon > 0$ be a real number. We want to prove that there exists some $J \in \mathbb{N}$ such that for all $j \geqslant J$, one has $|a_{n_j} - l| < \varepsilon$. Since $a_n \to l$, we know that there exists some $N \in \mathbb{N}$ such that for all $n \geqslant N$, one has $|a_n - l| < \varepsilon$. Noting (2.43), we see that $n_j \geqslant j$ for all $j \in \mathbb{N}$. Thus, set $J = N$, and note that whenever $j \geqslant J$, then $n_j \geqslant j \geqslant J = N$, and so we must have $|a_{n_j} - l| < \varepsilon$. □

*Lecture 15*

Here is an extremely powerful idea with a beautiful proof.

**Proposition 2.46.** *Any sequence of real numbers contains a monotonic subsequence.*

*Proof.* Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Let

$$S = \{n \in \mathbb{N} \ : \ a_n \geqslant a_k \ \text{ for all } \ k \geqslant n\}.$$

Now either the set $S$ has finitely many elements or the set $S$ is infinite.

If $S$ is infinite, then write $S$ in increasing order:

$$n_1 < n_2 < \cdots < n_j < n_{j+1} < \ldots$$

and consider the subsequence $(a_{n_j})_{j=1}^{\infty}$. We claim that this subsequence is decreasing. Indeed, note that for any $j \in \mathbb{N}$, since $n_j \in S$ and $n_{j+1} \geqslant n_j$, we must have $a_{n_j} > a_{n_{j+1}}$. Thus we are done in this case.

If $S$ is finite, let $m$ be the largest element of $S$. We will now inductively construct our increasing subsequence. Let $n_1 = m + 1$. Since $m + 1 \notin S$, there exists some $k \in \mathbb{N}$ with $k > m + 1$ such that $a_k > a_{m+1} = a_{n_1}$. Let $n_2 = k$. Now since $n_2 > n_1 = m + 1$, we must

have $n_2 \notin S$. Thus, there must exist some $k' \in \mathbb{N}$ such that $k' > n_2$ and $a_{k'} > a_{n_2}$. Let $n_3 = k'$. We keep repeating this to obtain a sequence of indices $(n_j)_{j=1}^{\infty}$ such that $a_{n_{j+1}} > a_{n_j}$ for all $j \in \mathbb{N}$. Thus, we are done in this case as well. □

As a straightforward corollary, we obtain the Bolzano–Weierstrass Theorem.

**Theorem 2.47** (Bolzano–Weierstrass Theorem)**.** *Any bounded sequence of real numbers contains a convergent subsequence.*

*Proof.* Let $(a_n)$ be this bounded sequence. Apply Proposition 2.46 to find a subsequence $(a_{n_j})_{j=1}^{\infty}$ which is monotonic. Since $(a_n)$ is bounded, it follows that $(a_{n_j})_{j=1}^{\infty}$ is also bounded. Now, if $(a_{n_j})_{j=1}^{\infty}$ is increasing, then we may apply Theorem 2.32 to deduce that $(a_{n_j})_{j=1}^{\infty}$ is convergent. On the other hand, if $(a_{n_j})_{j=1}^{\infty}$ is decreasing, then we may apply Corollary 2.35 to deduce that $(a_{n_j})_{j=1}^{\infty}$ is convergent. □

*Remark** 2.48. This is yet another statement that is equivalent to the least upper bound axiom.

This will allow us to prove the final result of this section.

**Theorem 2.49.** *A sequence of real numbers converges if and only if it is Cauchy.*

*Proof.* Lemma 2.40 already implies that any convergent sequence must be Cauchy, and so, it suffices to prove that any Cauchy sequence converges. Thus, let $(a_n)$ be some Cauchy sequence. Lemma 2.41 implies that $(a_n)$ is bounded, and so, Theorem 2.47 can be used to find some subsequence $(a_{n_j})_{j=1}^{\infty}$ and some $l \in \mathbb{R}$ such that $a_{n_j} \to l$ as $j \to \infty$.

Now, we just need to show that $a_n \to l$ as $n \to \infty$. Thus, letting $\varepsilon > 0$ be some real number, we need to find some $N \in \mathbb{N}$ such that for all $n \geqslant N$, we have $|a_n - l| < \varepsilon$. Since $a_{n_j} \to l$ as $j \to \infty$, there must exist some $J \in \mathbb{N}$ such that

$$|a_{n_j} - l| < \varepsilon/2 \tag{2.5}$$

for all $j \geqslant J$. Moreover, since $(a_n)$ is Cauchy, there exists some $N' \in \mathbb{N}$ such that for all $n, m \geqslant N'$, we have

$$|a_n - a_m| < \varepsilon/2. \tag{2.6}$$

Set $N = \max\{N', n_J\}$. This means that if $n_j \geqslant N$ for some $j \in \mathbb{N}$, then $n_j \geqslant n_J$ and so, we must have $j \geqslant J$. Thus, let $j \in \mathbb{N}$ be some natural number such that $n_j > N$. Note that for any $n \geqslant N$, we may apply the triangle inequality along with (2.5) and (2.6) to deduce that

$$|a_n - l| \leqslant |a_n - a_{n_j}| + |a_{n_j} - l| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

This concludes our proof of Theorem 2.49. □

## 3. Summation of series

3.1. **Motivation.** Given a sequence $(a_n)_{n=1}^{\infty}$, we would often be interested in the idea of an 'infinite sum' of the form $\sum_{n=1}^{\infty} a_n$.

Perhaps one has seen the following expression:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 2.$$

Indeed, if the left hand side was $S$, then we can perform some (non-rigorous) manipulations to get that

$$S = 2S - S = \left(2 + 1 + \frac{1}{2} + \frac{1}{4} + \dots\right) - \left(1 + \frac{1}{2} + \frac{1}{4} + \dots\right) = 2.$$

On the other hand, these (non-rigorous) manipulations do not work all the time. For instance, let

$$S' = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots.$$

Thus we have

$$\frac{S'}{2} = \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \cdots = 0 + \frac{1}{2} + 0 - \frac{1}{4} + 0 + \frac{1}{6} + 0 - \frac{1}{8} + \dots.$$

This means that

$$\frac{3S'}{2} = 1 + \frac{1}{3} - \left(\frac{1}{4} + \frac{1}{4}\right) + \frac{1}{5} + \frac{1}{7} - \left(\frac{1}{8} + \frac{1}{8}\right) + \cdots = 1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} \cdots = S'.$$

Thus we must have $S' = 0$. On the other hand, $S' \geqslant 1/2$ since $1/(2k-1) > 1/(2k)$ for all integers $k \geqslant 2$.

*So what has gone wrong? When can these manipulations be made rigorous?*

3.2. **Basic definitions and properties.** With the tools from the previous chapter, we can now provide a way to start answering these questions.

**Definition 3.1.** Given a sequence $(a_n)_{n=1}^{\infty}$ of real numbers, we define the sequence $(s_k)_{k=1}^{\infty}$ by setting

$$s_k = a_1 + \cdots + a_k = \sum_{i=1}^{k} a_i$$

for all $k \in \mathbb{N}$. We refer to $(s_k)$ as the sequence of *partial sums*.

Let $A$ be some real number. We say that

$$\sum_{n=1}^{\infty} a_n = A \quad \text{if} \quad s_k \to A \text{ as } k \to \infty. \tag{3.1}$$

In this case, we refer to $\sum_{n=1}^{\infty} a_n$ as a *series*.

**Example 3.2.** Let $r$ be a real number satisfying $|r| < 1$. Then

$$\sum_{n=1}^{\infty} r^n = \frac{r}{1-r}.$$

*Proof.* In order to see this, note that the partial sum $s_k$ satisfies

$$s_k = \sum_{i=1}^{k} r^i = \frac{r - r^{k+1}}{1-r};$$

indeed, one has

$$(1-r)s_k = \sum_{i=1}^{k} r^i - \sum_{i=1}^{k} r^{i+1} = r - r^{k+1}.$$

Applying algebra of limits and Example 2.30 part (2), we now see that

$$\lim_{k\to\infty} s_k = \frac{r}{1-r} - \lim_{k\to\infty} \frac{r^{k+1}}{1-r} = \frac{r}{1-r}.$$

Noting (3.1) now delivers the required conclusion. □

Similarly, we can show that

$$\sum_{n=1}^{\infty} \frac{9}{10^n} = 1.$$

**Example 3.3.** We have

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1.$$

*Proof.* Using the identity

$$\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1},$$

we see that for any $k \in \mathbb{N}$, one has

$$s_k = \sum_{n=1}^{k} \frac{1}{n(n+1)} = \sum_{n=1}^{k} \left( \frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{k+1}.$$

Thus, $\lim_{k\to\infty} s_k = 1$. □

We can extend algebra-of-limits style ideas to the setting of series as well.

**Lemma 3.4.** *Suppose $(a_n)$ and $(b_n)$ are sequences and $A, B$ are real numbers such that $A = \sum_{n=1}^{\infty} a_n$ and $B = \sum_{n=1}^{\infty} b_n$. Then*

$$\sum_{n=1}^{\infty} (a_n + b_n) = A + B. \tag{3.2}$$

*Moreover, for any $\lambda \in \mathbb{R}$, one has*

$$\sum_{n=1}^{\infty}(\lambda a_n) = \lambda A. \tag{3.3}$$

*Proof.* We first prove (3.2). The hypothesis implies that

$$\sum_{i=1}^{k} a_i \to A, \text{ as } k \to \infty \text{ and } \sum_{i=1}^{k} b_i \to B \text{ as } k \to \infty.$$

By algebra of limits, we deduce that

$$\sum_{i=1}^{k} a_i + \sum_{i=1}^{k} b_i = \sum_{i=1}^{k}(a_i + b_i) \to A + B.$$

This precisely means that (3.2) holds. One can prove (3.3) similarly by using algebra of limits on the partial sums. □

In the last section, we showed that if a sequence converges, then it must be bounded. We now prove the analogue of this result for series.

**Lemma 3.5.** *Suppose $(a_n)$ is a sequence and $A$ is some real number such that $\sum_{n=1}^{\infty} a_n = A$. Then $a_n \to 0$ as $n \to \infty$.*

*Proof.* For every $k \in \mathbb{N}$, let $s_k = \sum_{i=1}^{k} a_i$. By definition, we get that $s_k \to A$ as $k \to \infty$. On the other hand, for any $k \geqslant 2$, we have $a_k = s_k - s_{k-1}$, and so,

$$\lim_{k\to\infty} a_k = \lim_{k\to\infty}(s_k - s_{k-1}) = \lim_{k\to\infty} s_k - \lim_{k\to\infty} s_{k-1} = A - A = 0.$$

The second step above used algebra of limits while the penultimate step used the shift rule. □

*Lecture 17*

3.3. **Series with positive terms.** It is natural to wonder if the converse of Lemma 3.5 holds true. Unfortunately, one can show that this is not true.

**Example 3.6.** Let $a_n = 1/n$ for all $n \in \mathbb{N}$. Then $a_n \to 0$ as $n \to \infty$ but there is no real number $A \in \mathbb{R}$ such that $\sum_{n=1}^{\infty} a_n = A$.

*Proof.* It is obvious that $1/n \to 0$ as $n \to \infty$, and so, we prove the second part of the statement. In fact, we will prove something stronger:

$$s_k = \sum_{i=1}^{k} \frac{1}{i} \to \infty \quad \text{as} \quad k \to \infty. \tag{3.4}$$

Since $1/i > 0$ for all $i \in \mathbb{N}$, we see that $(s_k)_k$ is an increasing sequence, and so, by Theorem 2.32, it suffices to show that $(s_k)_k$ is not bounded above. In order to prove the latter statement, we will show that for any $j \in \mathbb{N}$, one has

$$s_{2^j} > j/2. \tag{3.5}$$

Indeed, this may then be combined with the Archimedean principle to deduce that $(s_k)_k$ is not bounded above.

We now prove (3.5). To do so, note that for any $j \in \mathbb{N}$, one has

$$\begin{aligned} s_{2^j} &= \sum_{i=1}^{2^j} \frac{1}{i} \geqslant 1 + \sum_{n=1}^{j} \sum_{i=2^{n-1}+1}^{2^n} \frac{1}{i} \geqslant 1 + \sum_{n=1}^{j} \frac{1}{2^n} \sum_{i=2^{n-1}+1}^{2^n} 1 \\ &= 1 + \sum_{n=1}^{j} \frac{1}{2^n} 2^{n-1} = 1 + \frac{j}{2} > j/2. \end{aligned}$$

□

We will now build further on Example 3.6.

**Definition 3.7.** Let $(a_n)$ be a sequence such that $a_n \geqslant 0$ for all $n \in \mathbb{N}$. Let $s_k = \sum_{i=1}^{k} a_i$ for all $k \in \mathbb{N}$. Note that $(s_k)$ is an increasing sequence, so by Theorem 2.32, either $s_k$ converges, or $s_k \to \infty$ as $k \to \infty$.

If $s_k \to \infty$ as $k \to \infty$, then we write

$$\sum_{n=1}^{\infty} a_n = \infty,$$

or that $\sum_n a_n$ *diverges*. Theorem 2.32 tells us that this is precisely when $(s_k)$ is not bounded above.

If $s_k$ converges, we write

$$\sum_{n=1}^{\infty} a_n < \infty.$$

Theorem 2.32 tells us that $s_k$ is convergent if and only if it is bounded.

**Example 3.8.** We will show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

*Proof.* Note that for any $n \geqslant 2$, one has $n^2 \geqslant n(n-1) > 0$. Hence, for any $k \in \mathbb{N}$, one has

$$\sum_{n=1}^{k} \frac{1}{n^2} \leqslant 1 + \sum_{n=2}^{k} \frac{1}{n(n-1)} = 1 + \sum_{n=2}^{k} \left( \frac{1}{n-1} - \frac{1}{n} \right) = 1 + 1 - \frac{1}{k} < 2.$$

Thus the sequence of partial sums is increasing and bounded above, and so, we are done. □

3.4. **Convergence test for series I.** The above type of proof generalises naturally to give the following 'test of convergence'.

**Lemma 3.9** (Comparison test for series)**.** *Suppose $N$ is some positive integer and suppose that $(a_n), (b_n)$ are sequences such that $0 \leqslant a_n \leqslant b_n$ for all $n \geqslant N$. Then*

$$\sum_{n=1}^{\infty} b_n < \infty \quad \Rightarrow \quad \sum_{n=1}^{\infty} a_n < \infty,$$

*and*

$$\sum_{n=1}^{\infty} a_n = \infty \quad \Rightarrow \quad \sum_{n=1}^{\infty} b_n = \infty.$$

*Proof.* We will first prove that we may assume $N = 1$. Indeed, note that whenever $k \geqslant N$, we have

$$\sum_{n=1}^{k} a_n = (a_1 + \cdots + a_{N-1}) + \sum_{n=N}^{k} a_n.$$

Since $N$ is some fixed positive integer, the sequence of partial sums $(\sum_{n=1}^{k} a_n)_{k=1}^{\infty}$ is bounded above if and only if the sequence of partial sums $(\sum_{n=1}^{k'} a_{n+N-1})_{k'=1}^{\infty}$ is bounded above.

Thus, from now on, we may assume that $N = 1$. In this case, let

$$s_k = \sum_{i=1}^{k} a_i \quad \text{and} \quad t_k = \sum_{i=1}^{k} b_i$$

for all $k \in \mathbb{N}$. The hypothesis (with $N = 1$) then implies that $0 \leqslant s_k \leqslant t_k$ for all $k \in \mathbb{N}$. Thus, if $(t_k)$ is bounded above, then $(s_k)$ must be bounded above. Similarly, if $s_k \to \infty$, then $t_k \to \infty$, see Lemma 2.28. □

*Remark* 3.10. This immediately combines with Example 3.8 to show that for any rational $p \geqslant 2$, one has

$$\sum_{n=1}^{\infty} \frac{1}{n^p} < \infty.$$

Similarly, noting Lemma 3.9 with (3.4) tells us that for any rational $p \leqslant 1$, one has

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \infty.$$

What about the case when $1 < p < 2$?

**Example 3.11.** Let

$$a_n = \frac{n+1}{n^3 - n - 1}$$

for all $n \in \mathbb{N}$. Note that this is sensible since $n^3 - n - 1 \neq 0$ for any $n \in \mathbb{N}$. We have $\sum_{n=1}^{\infty} a_n < \infty$.

*Proof.* Heuristically, we expect

$$a_n \approx \frac{n}{n^3} = \frac{1}{n^2},$$

and $\sum_{n=1}^{\infty} 1/n^2$ converges, so we may expect $\sum_{n=1}^{\infty} a_n$ converges.

We can make this more precise: note that for any $n \geqslant 10$, one has $n^3 - n - 1 \geqslant n^3/2$ and $n + 1 \leqslant 2n$. Thus, we get that

$$0 \leqslant a_n \leqslant \frac{2n}{n^3/2} = \frac{4}{n^2}$$

whenever $n \geqslant 10$. The fact that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges may be combined with Lemma 3.4 to deduce that $\sum_{n=1}^{\infty} 4/n^2$ converges. We may now apply Lemma 3.9 to deduce that $\sum_{n=1}^{\infty} a_n$ converges. □

**Example 3.12.** Let

$$a_n = \frac{n-2}{n^2 + 2n + 4}$$

for all $n \in \mathbb{N}$. Then $\sum_{n=1}^{\infty} a_n = \infty$.

*Proof.* As before, we heuristically expect that $a_n \approx n/n^2 = 1/n$ and so, $\sum_{n=1}^{\infty} a_n = \infty$.

In order to make this heuristic more concrete, note that whenever $n \geqslant 5$, we have $n - 2 \geqslant n/2$ and $n^2 + 2n + 4 \leqslant 4n^2$. Thus, whenever $n \geqslant 5$, we have

$$\frac{n-2}{n^2 + 2n + 4} \geqslant \frac{n/2}{4n^2} = \frac{1}{8n} > 0.$$

We note that $\sum_{n=1}^{\infty} 1/n = \infty$, and so, $\sum_{n=1}^{\infty} 1/(8n) = \infty$. Finally, Lemma 3.9 now implies that $\sum_{n=1}^{\infty} a_n = \infty$. □

*Lecture 18*

3.5. **The number** $e$**.** *[The material in §3.5 is non-examinable]*

Let

$$a_n = 1/(n-1)!$$

for every $n \in \mathbb{N}$. Note that for any $n \geqslant 3$, we have

$$(n-1)! \geqslant 2^{n-2} > 0, \quad \text{that is} \quad 0 < \frac{1}{(n-1)!} \leqslant \frac{1}{2^{n-2}},$$

and so, we may apply Lemma 3.9 to deduce that

$$\sum_{n=1}^{\infty} \frac{1}{(n-1)!} < \infty.$$

**Definition 3.13.** Let $e$ be the unique real number such that

$$\sum_{n=1}^{\infty} \frac{1}{(n-1)!} = e.$$

**Lemma 3.14.** *Let $k \geqslant 2$ be a positive integer. Then*

$$0 < \left| e - \sum_{n=1}^{k} \frac{1}{(n-1)!} \right| < \frac{1}{(k-1)!}.$$

*Proof.* We will first focus on the lower bound. By definition of convergence, we may find some $M \in \mathbb{N}$ such that whenever $m' \geqslant M$, one has

$$\left| e - \sum_{n=1}^{m'} \frac{1}{(n-1)!} \right| < \frac{1}{(k!)^2}.$$

Choosing $m' = \max\{M, k+10\}$, we may apply the triangle inequality to get that

$$\frac{1}{k!} < \left| \sum_{n=k+1}^{m'} \frac{1}{(n-1)!} \right| \leqslant \left| e - \sum_{n=1}^{m'} \frac{1}{(n-1)!} \right| + \left| e - \sum_{n=1}^{k} \frac{1}{(n-1)!} \right| < \frac{1}{(k!)^2} + \left| e - \sum_{n=1}^{k} \frac{1}{(n-1)!} \right|.$$

This means that

$$\left| e - \sum_{n=1}^{k} \frac{1}{(n-1)!} \right| > \frac{1}{k!} \left( 1 - \frac{1}{k!} \right).$$

We will now focus on the upper bound. Choosing $m'$ as before, we get that

$$\begin{aligned} \left| e - \sum_{n=1}^{k} \frac{1}{(n-1)!} \right| &\leqslant \left| e - \sum_{n=1}^{m'} \frac{1}{(n-1)!} \right| + \left| \sum_{n=k+1}^{m'} \frac{1}{(n-1)!} \right| \\ &\leqslant \frac{1}{(k!)^2} + \frac{1}{k!} \sum_{n=k+1}^{m'} \frac{k!}{(n-1)!} \leqslant \frac{1}{(k!)^2} + \frac{1}{k!} \left( 1 + \frac{1}{k} + \frac{1}{k^2} + \dots \right) \\ &\leqslant \frac{1}{(k!)^2} + \frac{1}{k!} \frac{k}{k-1} < \frac{1}{(k-1)!} \end{aligned}$$

□

We will use this to show that $e$ is irrational.

**Lemma 3.15.** *There exist no integers $p, q$ with $q \neq 0$ such that $e = p/q$.*

*Proof.* We prove this via contradiction and so, assume that $e = p/q$ such that $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Since $e > 1$, we must have $p \in \mathbb{N}$. Applying Lemma 3.14 with $k = q+1$, we see that

$$0 < \left| \frac{p}{q} - \sum_{n=1}^{q+1} \frac{1}{(n-1)!} \right| < \frac{1}{q!}.$$

Simplifying, we find that

$$0 < \left| p(q-1)! - \sum_{n=1}^{q+1} \frac{q!}{(n-1)!} \right| < 1.$$

Writing $m = p(q-1)! - \sum_{n=1}^{q+1} \frac{q!}{(n-1)!}$, we see that $m \in \mathbb{Z}$. On the other hand the above inequality implies that $0 < |m| < 1$, which can not happen if $m \in \mathbb{Z}$. Thus, we obtain the desired contradiction. □

*Remark* 3.16. Lemma 3.15 can be rephrased as follows: there exist no $p, q \in \mathbb{Z}$ such that $e$ is a solution to the linear equation $qx - p = 0$. But this notion can be generalised: do there exist $p_1, p_2, p_3 \in \mathbb{Z}$ such that $e$ is a solution to quadratic equation $p_1x^2 + p_2x + p_3 = 0$? What about higher degree equations?

It turns out $e$ is a *Transcendental number*, that is, there is no choice of $d, p_1, \ldots, p_{d+1} \in \mathbb{Z}$ with $d \geqslant 1$ such that $e$ is a solution to the equation $p_1x^d + p_2x^{d-1} + \cdots + p_dx + p_{d+1} = 0$.

We will now prove an estimate that we claimed in the first lecture.

**Lemma 3.17.** *We have*

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e.$$

*Proof.* Using binomial theorem, we proved that for any $n \in \mathbb{N}$, one has

$$\left(1 + \frac{1}{n}\right)^n \leqslant 1 + \sum_{j=1}^{n} \frac{1}{j!},$$

see (1.1). In the same example, we further showed that $\left(1 + \frac{1}{n}\right)^n \to l$ for some $l \in \mathbb{R}$. Since limits preserve inequalities, the preceding inequality gives us

$$l \leqslant e.$$

We will now show that $l \geqslant e$. It suffices to prove that for any $k \in \mathbb{N}$, one has

$$l \geqslant \sum_{n=1}^{k+1} \frac{1}{(n-1)!},$$

since then, we can again use the fact that limits preserve inequalities to get that $l \geqslant e$. Thus, let $k \in \mathbb{N}$ be some fixed integer. Note that, for every $m \in \mathbb{N}$, one has

$$\left(1 + \frac{1}{m}\right)^{k+m} = \sum_{j=0}^{k+m} \binom{k+m}{j} \frac{1}{m^j} \geqslant \sum_{j=0}^{k} \frac{1}{j!} \frac{(m+k)(m+k-1)\ldots(m+k-j+1)}{m^j} = \sum_{j=0}^{k} \frac{1}{j!}.$$

Now, taking limits as $m \to \infty$, we see that

$$\lim_{m\to\infty} \left(1+\frac{1}{m}\right)^k \left(1+\frac{1}{m}\right)^m \geqslant \sum_{j=0}^{k} \frac{1}{j!}.$$

On the other hand, applying algebra of limits, the left hand side is

$$\left(1+\lim_{m\to\infty}\frac{1}{m}\right)^k \lim_{m\to\infty}\left(1+\frac{1}{m}\right)^m = 1^k \cdot l = l.$$

This is precisely the claimed estimate. □

*Lecture 19*

3.6. **Absolute and conditional convergence.** We now return to the realm of general sequences $(a_n)$, where $a_n$ is some arbitrary real number for all $n \in \mathbb{N}$.

**Definition 3.18.** We say that a series $\sum_{n=1}^{\infty} a_n$ *converges absolutely* if $\sum_{n=1}^{\infty} |a_n| < \infty$.

There are series that converge but do not converge absolutely.

**Example 3.19.** Let $a_n = (-1)^{n+1}/n$ for all $n \in \mathbb{N}$. Then

$$\sum_{n=1}^{\infty} a_n < \infty \quad \text{but} \quad \sum_{n=1}^{\infty} |a_n| = \infty.$$

*Proof.* Note that $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} 1/n = \infty$. On the other hand, for any $k \in \mathbb{N}$, we have

$$0 < s_{2k} = \sum_{n=1}^{2k} \frac{(-1)^{n+1}}{n} = \sum_{n'=1}^{k} \left(\frac{1}{2n'-1} - \frac{1}{2n'}\right) = \sum_{n'=1}^{k} \frac{1}{2n'(2n'-1)} \leqslant \sum_{n'=1}^{k} \frac{1}{n'^2}.$$

Since $\sum_{n=1}^{\infty} 1/n^2 < \infty$, we get that $(s_{2k})_{k=1}^{\infty}$ is an increasing sequence which is bounded above, and so, $(s_{2k})_k$ converges. Moreover, $s_{2k+1} = s_{2k} + 1/(2k+1)$, and so, $\lim_{k\to\infty} s_{2k+1} = \lim_{k\to\infty} s_{2k}$. One can now use the definition of convergence to show that $\lim_{k\to\infty} s_k = \lim_{k\to\infty} s_{2k}$, which in turn means that $\sum_{n=1}^{\infty} a_n < \infty$. □

**Definition 3.20.** We say that a series $\sum_{n=1}^{\infty} a_n$ *converges conditionally* if $\sum_{n=1}^{\infty} |a_n| = \infty$ but $\sum_{n=1}^{\infty} a_n$ converges.

On the other hand, if a series converges absolutely, then it must converge as well.

**Lemma 3.21.** *Suppose* $\sum_{n=1}^{\infty} |a_n| < \infty$. *Then* $\sum_{n=1}^{\infty} a_n < \infty$.

*Proof.* We will define two auxiliary sequences. Given $n \in \mathbb{N}$, if $a_n \geqslant 0$, then define $b_n = a_n$ and $c_n = 0$, while if $a_n < 0$, then define $b_n = 0$ and $c_n = -a_n$. In particular, we see that

$a_n = b_n - c_n$ for all $n \in \mathbb{N}$. Moreover, we have $0 \leqslant b_n, c_n \leqslant |a_n|$ for all $n \in \mathbb{N}$. It follows from comparison test that

$$\sum_{n=1}^{\infty} b_n < \infty \quad \text{and} \quad \sum_{n=1}^{\infty} c_n < \infty.$$

Finally, since $a_n = b_n - c_n$, we may apply Lemma 3.4 to deduce that $\sum_{n=1}^{\infty} a_n$ converges. □

3.7. **Convergence test for series II.** Lemma 3.21 and Lemma 3.9 combine to deliver the following comparison test.

**Lemma 3.22.** *Let $(a_n), (b_n)$ be sequences such that $|a_n| \leqslant b_n$ for all $n \in \mathbb{N}$. Moreover, suppose that $\sum_{n=1}^{\infty} b_n < \infty$. Then $\sum_{n=1}^{\infty} a_n$ converges.*

Here is an analogue of the ratio test for series.

**Lemma 3.23** (Ratio test for series)**.** *Suppose $(a_n)$ is a sequence such that $a_n \neq 0$ for all $n \in \mathbb{N}$. Moreover, suppose that $r \in \mathbb{R}$ satisfies*

$$\left| \frac{a_{n+1}}{a_n} \right| \to r \quad \text{as} \quad n \to \infty.$$

*Then*

*(1) if $r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges absolutely;*
*(2) if $r > 1$, the series $\sum_{n=1}^{\infty} a_n$ does not converge;*
*(3) if $r = 1$, it could be either.*

*Proof.* We will proceed case by case.

(1) We will first prove (1). The hypothesis implies that there exists some $N \in \mathbb{N}$ such that

$$\frac{|a_{n+1}|}{|a_n|} \leqslant \frac{1+r}{2} \tag{3.6}$$

holds for all $n \geqslant N$. In particular, this means that for all $n \geqslant N$, one has

$$|a_n| \leqslant p^{n-N} |a_N|$$

where $p = (1+r)/2 < 1$. Since $0 < p < 1$, we have that $\sum_{n=1}^{\infty} p^n < \infty$, in which case, we may apply comparison test (Lemma 3.9) to deduce that $\sum_{n=1}^{\infty} |a_n|$ converges.

(2) As in the preceding case, we get that there exists some $N \in \mathbb{N}$ such that for all $n \geqslant N$, one has

$$|a_n| \geqslant \left( \frac{r+1}{2} \right)^{n-N} |a_N|. \tag{3.7}$$

Since $|a_N| \neq 0$ and $(r+1)/2 > 1$, we see that the right hand side $\to \infty$ as $n \to \infty$. In particular, $\sum_{n=1}^{\infty} a_n$ can not converge, since if it did, Lemma 3.5 would imply that

$|a_n| \to 0$ as $n \to \infty$.

(3) We now consider (3). Note that the series $\sum_{n=1}^{\infty} 1/n = \infty$ while $\sum_{n=1}^{\infty} 1/n^2 < \infty$, but both of the corresponding ratios $|a_{n+1}/a_n| \to 1$ as $n \to \infty$. □

*Lecture 20*

*Remark* 3.24. Note that the above proofs only require (3.6) and (3.7) to hold. So even if one can not explicitly calculate the limit $|a_{n+1}/a_n|$, sometimes it suffices to have the corresponding lower or upper bounds.

For instance, suppose one has a series $\sum_{n=1}^{\infty} a_n$ and some $N \in \mathbb{N}$ such that $|a_{n+1}/a_n| \leqslant 1/2$ for all $n \geqslant N$. Then as in the above proof, we can show that whenever $n \geqslant N$, then one has

$$|a_n| \leqslant |a_{n-1}| \left(\frac{1}{2}\right) \leqslant |a_{n-2}| \left(\frac{1}{2}\right)^2 \leqslant \ldots \leqslant |a_{N+1}| \left(\frac{1}{2}\right)^{n-N-1} \leqslant |a_N| \left(\frac{1}{2}\right)^{n-N}.$$

Since $N$ is fixed, we get that $|a_N| 2^N \sum_{n=1}^{\infty} \frac{1}{2^n} < \infty$ and so by comparison test, we deduce that $\sum_{n=1}^{\infty} |a_n| < \infty$.

**Example 3.25.** Given some $x \in \mathbb{R}$, define the sequence $(a_n)$ as $a_n = x^n/n!$ for all $n \in \mathbb{N}$. If $x = 0$, then $a_n = 0$ for all $n \in \mathbb{N}$, in which case $\sum_{n=1}^{\infty} |a_n| < \infty$ trivially.

On the other hand, if $x \neq 0$, then $a_n \neq 0$ for all $n \in \mathbb{N}$. In this case, we have

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{x}{n+1}\right| \to 0 \quad \text{as} \quad n \to \infty$$

for any fixed $x \in \mathbb{R}$. The ratio test now tells us that the series $\sum_{n=1}^{\infty} \frac{x^n}{n!}$ converges absolutely for any $x \in \mathbb{R}$.

**Example 3.26.** Define the sequence $(a_n)$ as $a_n = n!(2n)!/(3n)!$ for all $n \in \mathbb{N}$. Note that $a_n \neq 0$ for all $n \in \mathbb{N}$. Moreover, we get that

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{(n+1)(2n+1)(2n+2)}{(3n+1)(3n+2)(3n+3)} = \frac{4}{27} < 1.$$

Thus the ratio test implies that $\sum_{n=1}^{\infty} |a_n| < \infty$.

**Lemma 3.27** (Cauchy's root test)**.** *Suppose that $(a_n)$ is a sequence such that $|a_n|^{1/n} \to r$ as $n \to \infty$. Then*

*(1) if $r < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges absolutely;*
*(2) if $r > 1$, the series $\sum_{n=1}^{\infty} a_n$ does not converge;*
*(3) if $r = 1$, it could be either.*

*Proof.* The proof is very similar to the proof of the ratio test. When we are in the setting of (1), we can show that there exists some $N \in \mathbb{N}$ such that $|a_n| \leqslant p^n$ for all $n \geqslant N$, for some absolute constant $p < 1$. Comparison test now gives us the desired conclusion. Similarly, we can show that in the setting of (2), one has $|a_n| \geqslant p^n$ for all $n \geqslant N$, for some absolute constant $p > 1$, which can then be combined with Lemma 3.5 to obtain the claimed conclusion. Finally, the justification behind (3) is the same as that of (3) in Lemma 3.23. □

**Example 3.28.** Define the sequence $(a_n)$ as

$$a_n = \left(\frac{n^3 + 10}{2n^3 - 3}\right)^n$$

for all $n \in \mathbb{N}$. Then $\sum_{n=1}^{\infty} a_n$ converges.

*Proof.* Note that

$$\lim_{n\to\infty} |a_n|^{1/n} = \lim_{n\to\infty} \left|\frac{n^3 + 10}{2n^3 - 3}\right| = \lim_{n\to\infty} \frac{1 + 10/n^3}{2 - 3/n^3} = \frac{1}{2}.$$

Thus by the root test, we see that $\sum_{n=1}^{\infty} a_n$ converges. □

3.8. **Integral test.** Things we have not yet rigorously covered: given a function $f : \mathbb{R} \to \mathbb{R}$, what does the "integral" $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x$ mean? Which functions $f$ can be "integrated"? These will be covered in a later module.

On the other hand, one can often bound certain series by integrals and it is an extremely useful trick to test convergence/divergence of series.

**Definition 3.29.** We say that a function $f : [1, \infty) \to [0, \infty)$ is *decreasing* if for any $1 \leqslant x \leqslant y$, we have $f(x) \geqslant (y)$.

**Lemma 3.30** (Integral test)**.** *Suppose that $f : [1, \infty) \to [0, \infty)$ is a decreasing function. Then*

*(1) If $\int_1^k f(x)\,\mathrm{d}x$ is bounded as $k \to \infty$, then $\sum_{n=1}^{\infty} f(n)$ converges;*
*(2) If $\int_1^k f(x)\,\mathrm{d}x$ is unbounded as $k \to \infty$, then $\sum_{n=1}^{\infty} f(n)$ diverges.*

*Proof.* Since $f$ is decreasing, for any integer $n \geqslant 2$, we have that $f(n) \leqslant f(x)$ for all $x \in (n-1, n]$. This implies that

$$f(n) \leqslant \int_{n-1}^{n} f(x)\,\mathrm{d}x.$$

Similarly, we get that

$$\int_{n}^{n+1} f(x)\,\mathrm{d}x \leqslant f(n)$$

for all $n \in \mathbb{N}$. Let $b_1 = 0$ and let $b_{n+1} = \int_{n-1}^{n} f(x)\ \mathrm{d}x$ for all $n \in \mathbb{N}$. Note that

$$\sum_{n=1}^{k} b_n = \sum_{n=2}^{k} \int_{n-1}^{n} f(x)\ \mathrm{d}x = \int_{1}^{k} f(x)\ \mathrm{d}x.$$

The preceding inequalities imply that

$$0 \leqslant b_{n+1} \leqslant f(n) \leqslant b_n$$

for all integers $n \geqslant 2$. We can now apply comparison test (Lemma 3.9) to deduce the desired conclusion. Indeed, if

$$\sum_{n=1}^{k} b_n = \int_{1}^{k} f(x)\ \mathrm{d}x \to \infty$$

as $k \to \infty$, then $\sum_n b_{n+1} = \infty$, and so, the comparison test tells us that $\sum_n f(n) = \infty$. Similarly, if

$$\sum_{n=1}^{k} b_n = \int_{1}^{k} f(x)\ \mathrm{d}x$$

is bounded above, then the comparison test implies that $\sum_n f(n) < \infty$. □

**Example 3.31.** In Remark 3.10, we saw that whenever $p \in \mathbb{Q}$ satisfies $p \geqslant 2$, then $\sum_n 1/n^p < \infty$ while if $p \leqslant 1$, then $\sum_n 1/n^p = \infty$. We can now cover the range $p \in (1, 2)$ using the integral test.

Indeed, note that the function $f(x) = 1/x^p$ is positive and decreasing when $x \geqslant 1$ and $p > 1$. Moreover,

$$\int_{1}^{k} f(x)\ \mathrm{d}x = \int_{1}^{k} 1/x^p\ \mathrm{d}x = \frac{1}{p-1}\left(1 - \frac{1}{k^{p-1}}\right) < \frac{1}{p-1}$$

for all integers $k \geqslant 2$. The integral test then tells us that $\sum_n 1/n^p$ converges for all $p > 1$.

Similarly, one can apply the integral test to deduce that $\sum_n 1/n^p$ diverges when $p \in (0, 1]$.

*Remark** 3.32. Purists can deduce the above conclusion without using the integral test. In particular, let $p \in \mathbb{Q}$ satisfy $p \in (1, 2)$, and for any $k \in \mathbb{N}$, let $s_k = \sum_{n=1}^{k} 1/n^p$. Since $(s_k)_k$ is increasing, it suffices to prove that $(s_{2^n-1})_n$ is bounded above. For any integer $n \geqslant 2$, note that

$$s_{2^n-1} = \sum_{j=1}^{n} \sum_{2^{j-1} \leqslant n < 2^j} \frac{1}{n^p} \leqslant \sum_{j=1}^{n} \frac{1}{(2^{j-1})^p} \sum_{2^{j-1} \leqslant n < 2^j} 1 = \sum_{j=1}^{n} \frac{2^{j-1}}{2^{(j-1)p}}.$$

Since $p > 1$, we have that $2^{p-1} > 1$, and so, we get that

$$s_{2^n-1} \leqslant \sum_{j=1}^{n} \frac{1}{(2^{p-1})^{j-1}} \leqslant \frac{1}{1 - \frac{1}{2^{p-1}}}.$$

Thus we have shown that $(s_{2^n-1})_n$ is bounded above, and since $(s_k)_k$ is increasing, this implies that $(s_k)_k$ is also bounded above. This means that $s_k$ converges as $k \to \infty$, that is, $\sum_n 1/n^p < \infty$.

*Lecture 21*

As we previously mentioned, there are series that converge but do not converge absolutely, that is, they converge conditionally.

### 3.9. **Alternating series test.**

**Lemma 3.33.** *Suppose that $(a_n)$ is a sequence such that*

*(1) $a_n \geqslant 0$ for all $n \in \mathbb{N}$*
*(2) $a_n \geqslant a_{n+1}$ for all $n \in \mathbb{N}$*
*(3) $a_n \to 0$ .*

*Then*

$$\sum_{n=1}^{\infty}(-1)^{n+1}a_n$$

*converges.*

*Proof.* For any $k \in \mathbb{N}$, let $s_k = \sum_{n=1}^{k}(-1)^{n+1}a_n$. Then (2) implies that $s_{2k} \leqslant s_{2k+2}$ for any $k \in \mathbb{N}$. Noting (2), we get that $a_{2i} - a_{2i+1}$ for every $i \in \mathbb{N}$, and so,

$$s_{2k} = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots - (a_{2k-2} - a_{2k-1}) - a_{2k} \leqslant a_1 - a_{2k}.$$

We now use (1) to deduce that in fact $s_{2k} \leqslant a_1$ for all $k \in \mathbb{N}$. Thus $(s_{2k})_{k=1}^{\infty}$ is increasing and bounded above, and so, it must converge. Suppose $s_{2k} \to l$ as $k \to \infty$ for some $l \in \mathbb{R}$. Since $s_{2k+1} = s_{2k} + a_{2k+1}$, we use algebra of limits and (3) to deduce that $s_{2k+1} \to l$ as $k \to \infty$. We may now use the definition of convergence to show that $s_k \to l$ as $k \to \infty$. □

*Remark** 3.34. This test is a *sufficient* criterion but not a *necessary* criterion. This means that there are conditionally convergent series which do not satisfy the hypotheses of Lemma 3.33.

For example, consider

$$a_n = \frac{1}{n + (-1)^{n+1}}.$$

Note that $a_n > 0$ for all $n \in \mathbb{N}$, but $a_n$ is not decreasing. On the other hand, one can show that $\sum_n (-1)^{n+1}a_n$ converges.

3.10. **Rearrangements.** *[The material in §3.10 is non-examinable]*

Our aim in this section is to address the following "paradox": Let $S \in \mathbb{R}$ satisfy

$$S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}.$$

Then we may perform some rearrangements to see that

$$\begin{aligned} S &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \frac{1}{10} + \dots \\ &= \left(1 - \frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{6} - \frac{1}{8}\right) + \left(\frac{1}{5} - \frac{1}{10} - \frac{1}{12}\right) + \left(\frac{1}{7} - \frac{1}{14} - \frac{1}{16}\right) + \dots \\ &= \left(\frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{6} - \frac{1}{8}\right) + \left(\frac{1}{10} - \frac{1}{12}\right) + \left(\frac{1}{14} - \frac{1}{16}\right) + \dots \\ &= \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \dots\right) = \frac{S}{2}. \end{aligned}$$

Thus we must have $S = 0$.

On the other hand, we clearly have $S \geqslant 1/2$. Indeed, since

$$1 - \frac{1}{2} = \frac{1}{2} \quad \text{and} \quad \frac{1}{2l+1} - \frac{1}{2l+2} > 0$$

for all $l \in \mathbb{N}$, the partial sums $s_k = \sum_{n=1}^{k} (-1)^{n+1}/n \geqslant 1/2$ for all $k \geqslant 3$, and so, we may apply the fact that limits preserve non-strict inequalities to deduce that $S \geqslant 1/2$.

*So what has gone wrong?* Note that the series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots \quad \text{and} \quad \left(1 - \frac{1}{2} - \frac{1}{4}\right) + \left(\frac{1}{3} - \frac{1}{6} - \frac{1}{8}\right) + \left(\frac{1}{5} - \frac{1}{10} - \frac{1}{12}\right) + \dots$$

have very different partial sums. Thus doing weird rearrangements genuinely distorts the partial sums and subsequently changes the limit of the series.

In order to talk about this rigorously, we present some definitions.

**Definition 3.35.** Let $(a_n)$ be a sequence and let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection, id est, $\sigma$ is one-to-one (that is if $i, j \in \mathbb{N}$ satisfy $\sigma(i) = \sigma(j)$, then $i = j$) and onto (that is, for any $l \in \mathbb{N}$, there exists some $n \in \mathbb{N}$ such that $\sigma(n) = l$).

Given this bijection $\sigma$, a *rearrangement* of the sequence $(a_n)_n$ is $(a_{\sigma(n)})_n$.

Here are two nice results about rearranging series.

**Theorem 3.36.** *Let $(a_n)$ be a sequence such that $\sum_{n=1}^{\infty} |a_n| < \infty$. Thus, let $l = \sum_{n=1}^{\infty} a_n$. Then for all $\sigma : \mathbb{N} \to \mathbb{N}$, one has $\sum_{n=1}^{\infty} a_{\sigma(n)} = l$.*

**Theorem 3.37** (Riemann Rearrangement theorem)**.** *Let $(a_n)$ be a sequence such that $\sum_{n=1}^{\infty} a_n$ is conditionally convergent. Then for any $l \in \mathbb{R}$, there exists some bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that $\sum_{n=1}^{\infty} a_{\sigma(n)} = l$.*

Thus, any rearrangement of an absolutely convergent series still converges to the same limit, while a conditionally convergent series can be rearranged to converge to anything.

We will only prove Theorem 3.36.

*Proof of Theorem 3.36.* We begin by analysing the case when $a_n \geqslant 0$ for all $n \in \mathbb{N}$.

Let $b_n = a_{\sigma(n)}$ for all $n \in \mathbb{N}$. Our first aim is to show that $\sum_n b_n < \infty$. This is not hard to show: for any $N \in \mathbb{N}$, let

$$M_N = \max\{\sigma(n) : n \in \{1, 2, \ldots, N\}\}.$$

With this in hand, we note that since $a_n \geqslant 0$ for all $n \in \mathbb{N}$, one has

$$\sum_{n=1}^{k} b_n \leqslant \sum_{n=1}^{M_k} a_n. \tag{3.8}$$

Since $\sum_n a_n < \infty$, there exists some $L > 0$ such that the right hand side is bounded above by $L$, and so, $\sum_{n=1}^{k} b_n \leqslant L$ for all $k \in \mathbb{N}$. Since $b_n \geqslant 0$ for all $n \in \mathbb{N}$, we get that $\sum_n b_n$ converges.

We will now show that $\sum_n b_n = l$. The key point here is to note that $M_N \geqslant N$ for all $N \in \mathbb{N}$ (this follows from the fact that $\sigma : \mathbb{N} \to \mathbb{N}$ is one-to-one). Now, define for every $N \in \mathbb{N}$, the quantity

$$M'_N = \max\{\sigma^{-1}(n) : n \in \{1, 2, \ldots, N\}\}.$$

As before, we have that $M'_N \geqslant N$ for all $N \in \mathbb{N}$. Moreover, note that

$$\sum_{n=1}^{k} a_n \leqslant \sum_{n=1}^{M'_k} b_n \leqslant \sum_{n=1}^{M_{M'_k}} a_n$$

holds for any $k \in \mathbb{N}$, with the second inequality following from (3.8). As $k \to \infty$, we have $M'_k \to \infty$ and consequently, $M_{M'_k} \to \infty$. Furthermore, the series on the left and right hand side converge to $l$, whence, by the sandwich theorem, one has $\sum_n b_n = l$.

Thus, we have proven Theorem 3.36 if $a_n \geqslant 0$, and so, we now proceed with the general case when $a_n \in \mathbb{R}$. We define the sequences $(c_n)$ and $(d_n)$ as $c_n = a_n$ when $a_n \geqslant 0$ and $c_n = 0$ otherwise, and $d_n = c_n - a_n$. Note that $0 \leqslant c_n, d_n \leqslant |a_n|$ for all $n \in \mathbb{N}$. Thus, $\sum_n c_n < \infty$ and $\sum_n d_n < \infty$. By the preceding case, we see that

$$\sum_n c_{\sigma(n)} = \sum_n c_n \quad \text{and} \quad \sum_n d_{\sigma(n)} = \sum_n d_n.$$

This implies that

$$\sum_n a_{\sigma(n)} = \sum_n (c_{\sigma(n)} - d_{\sigma(n)}) = \sum_n c_{\sigma(n)} - \sum_n d_{\sigma(n)}$$
$$= \sum_n c_n - \sum_n d_n = \sum_n (c_n - d_n) = \sum_n a_n. \qquad \square$$

We will now prove the Riemann rearrangement theorem.

*Proof of Theorem 3.37.* Let $(c_n)$ be the sub-sequence of all non-negative elements in $(a_n)$ and let $(d_n)$ be the sub-sequence of all negative elements in $(a_n)$. In particular, for any $k \in \mathbb{N}$, there exist $k_1, k_2 \in \mathbb{N}$ such that $k_1 + k_2 = k$ and

$$\sum_{n \leqslant k_1} c_n + \sum_{n \leqslant k_2} d_n = \sum_{n \leqslant k} a_n \quad \text{and} \quad \sum_{n \leqslant k_1} |c_n| + \sum_{n \leqslant k_2} |d_n| = \sum_{n \leqslant k} |a_n|.$$

*Claim* 1. We claim that

$$\sum_{n=1}^{\infty} c_n = \infty \quad \text{and} \quad \sum_{n=1}^{\infty} d_n = -\infty.$$

In order to prove Claim 1, let $R > 0$, and note that since $\sum_n |a_n| = \infty$, there exists some $K \in \mathbb{N}$ such that for any $k \geqslant K$, one has

$$\sum_{n=1}^{k} |a_n| = \sum_{n=1}^{k_1} c_n - \sum_{n=1}^{k_2} d_n > R. \tag{3.9}$$

Moreover, since $\sum_n a_n = a$ for some $a \in \mathbb{R}$, there exists some $K' \in \mathbb{N}$ such that whenever $k \geqslant K'$, one has

$$\sum_{n=1}^{k} a_n = \sum_{n=1}^{k_1} c_n + \sum_{n=1}^{k_2} d_n \in (a/2, 3a/2). \tag{3.10}$$

(3.9) and (3.9) can be combined to deduce that whenever $k \geqslant \max\{K, K'\}$, one has

$$\sum_{n=1}^{k} c_n \geqslant \sum_{n=1}^{k_1} c_n > \frac{R}{2} + \frac{a}{4}, \quad \text{and} \quad \sum_{n=1}^{k} d_n \leqslant \sum_{n=1}^{k_2} d_n < \frac{3a}{4} - \frac{R}{2}.$$

Since $a$ is some fixed real number, we see that $\sum_n c_n = \infty$ and $\sum_n d_n = -\infty$.

*Claim* 2. We claim that $c_n, d_n \to 0$ as $n \to \infty$. Indeed, note that $|c_n|, |d_n| \leqslant |a_n|$ for all $n \in \mathbb{N}$, and so, since $\sum_{n=1}^{\infty} a_n < \infty$, we get that $a_n \to 0$ as $n \to \infty$, which in turn implies that $c_n, d_n \to 0$ as $n \to \infty$.

With these two claims in hand, we will now prove Theorem 3.37. In particular, we will prove the case when $l > 0$, the case when $l < 0$ and $l = 0$ follow similarly.

Since $\sum_n c_n \to \infty$, we know that there exists some $N'$ such that for all $k \geqslant N'$, one has $\sum_{n \leqslant k} c_n > l$. Let $N_1$ be the smallest such positive integer $N'$, that is,

$$\sum_{n \leqslant N_1} c_n > l \text{ but } \sum_{n \leqslant N_1 - 1} c_n \leqslant l.$$

This in fact implies that

$$l < \sum_{n \leqslant N_1} c_n \leqslant l + c_{N_1}.$$

Now, since $\sum_n d_n \to -\infty$, we may find some $M' \in \mathbb{N}$ such that whenever $k \geqslant M'$, one has

$$\sum_{n \leqslant k} d_n < l - \sum_{n \leqslant N_1} c_n.$$

Let $M_1$ be the smallest such positive integer. As before, this implies that

$$l + d_{M_1} \leqslant \sum_{n \leqslant N_1} c_n + \sum_{n \leqslant M_1} d_n < l.$$

We can now find some positive integer $N_2 > N_1$ such that

$$l < \sum_{n \leqslant N_2} c_n + \sum_{n \leqslant M_1} d_n \leqslant l + c_{N_2},$$

and then find some positive integer $M_2 > M_1$ such that

$$l + d_{M_2} \leqslant \sum_{n \leqslant N_2} c_n + \sum_{n \leqslant M_1} d_n < l,$$

and so on.

Thus our rearrangement of $(a_n)_n$ is:

$$c_1, c_2, \ldots, c_{N_1}, d_1, d_2, \ldots, d_{M_1}, c_{N_1+1}, c_{N_1+2}, \ldots, c_{N_2}, d_{M_1+1}, d_{M_1+2}, \ldots, d_{M_2}, c_{N_2+1}, \ldots, c_{N_3}, \ldots.$$

Note that for any $\varepsilon \in \mathbb{N}$, since $c_n, d_n \to 0$, one can find some sufficiently large $N$ such that for all $n \geqslant N$, one has $-\varepsilon < d_n < 0 \leqslant c_n \leqslant \varepsilon$. Noting our rearrangement, we deduce that there exists some sufficiently large $K \in \mathbb{N}$ such that any partial sum of the form $\sum_{n \leqslant k} a_{\sigma(n)}$, with $k \geqslant K$, can be shown to be bounded between

$$[l + d_{M_r}, l + c_{N_r}]$$

for some $r \geqslant N$. On the other hand, this means that this partial sum lies in the interval $(l - \varepsilon, l + \varepsilon)$. This concludes our proof. $\square$

## 4. Continuous functions

4.1. **Introduction.** Let $E \subseteq \mathbb{R}$ be a non-empty set and let $f : E \to \mathbb{R}$ be a function. We say that $E$ is the *domain* of $f$ and the set

$$f(E) = \{f(x) : x \in E\}$$

is called the *range* or *image* of $f$.

**Example 4.1.** Here are some examples:

(1) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ satisfy $f(x) = 1/x$ for all $x \in \mathbb{R} \setminus \{0\}$.
(2) Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x) = x^{10}$ for all $x \in \mathbb{R}$.
(3) Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x) = \lfloor x \rfloor$ for all $x \in \mathbb{R}$.
(4) Thomae's function $f : (0, 1) \to \mathbb{R}$ can be defined as

$$f(x) = \begin{cases} 1/q & x = p/q \text{ where } p, q \in \mathbb{N} \text{ are co-prime} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

(5) Let $f : [2, 100] \to \mathbb{R}$ satisfy $f(x) = 1/100$ for all $x \in [2, 100]$.

**Definition 4.2.** Let $E \subseteq \mathbb{R}$ be a non-empty set, let $f : E \to \mathbb{R}$ be a function, let $c \in E$ be an element. We say that $f$ is *continuous* at $c \in E$ if for all $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $x \in E$ satisfies $|x - c| < \delta$ we have $|f(x) - f(c)| < \varepsilon$.

We say that $f$ is *continuous on* $E$ if $f$ is continuous at $c$ for all $c \in E$. Equivalently, we may write that $f : E \to \mathbb{R}$ is *continuous*.

It is worth recalling that $x \in E$ satisfying $|x - c| < \delta$ is equivalent to $x \in E \cap (c - \delta, c + \delta)$.

**Example 4.3.** The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 100$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R}$.

*Proof.* We have to show that $f$ is continuous at $c$, for any $c \in \mathbb{R}$. Thus, let $c \in \mathbb{R}$ and let $\varepsilon > 0$ be given. With these fixed, choose $\delta = 1$ and note that for any $x \in (c - \delta, c + \delta)$, we have that

$$|f(x) - f(c)| = 0 < \varepsilon.$$

Thus $f$ is continuous at $c$. Since this works for all $c \in \mathbb{R}$, we get that $f$ is continuous on $\mathbb{R}$. □

In the above example, it turned out that our choice of $\delta$ was independent of $\varepsilon$ and $c$, but this is not always the case, as we see from the following two examples.

**Example 4.4.** The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x$ is continuous.

*Proof.* As before, let $c \in \mathbb{R}$ and $\varepsilon > 0$ be given. Choosing $\delta = \varepsilon$, we see that whenever $x \in (c - \delta, c + \delta)$, we have that

$$|f(x) - f(c)| = |x - c| < \delta = \varepsilon,$$

and so, $f$ is continuous at $c$. As in the preceding example, since this works for all $c \in \mathbb{R}$, we get that $f$ is continuous on $\mathbb{R}$. □

**Example 4.5.** The function $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by $f(x) = 1/x$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R} \setminus \{0\}$.

*Proof.* Let $c \in \mathbb{R} \setminus \{0\}$ and $\varepsilon > 0$ be given. We need to find some $\delta > 0$, such that whenever $x \in (c - \delta, c + \delta) \cap (\mathbb{R} \setminus \{0\})$, we get that

$$|f(x) - f(c)| = |1/x - 1/c| < \varepsilon.$$

Simplifying the expression in the middle, we see that

$$|1/x - 1/c| = |c - x|/|x||c|$$

whenever $x \neq 0$.

So we need to choose $\delta > 0$ such that whenever $x \in (c - \delta, c + \delta)$, we have

$$x \neq 0 \quad \text{and} \quad |x - c| < |x||c|\varepsilon. \tag{4.1}$$

If $\delta < |c|/2$, then for any $x$ satisfying $|x - c| < \delta$, we can apply the triangle inequality to deduce that

$$|x| \geqslant |c| - |x - c| > |c|/2 > 0.$$

Moreover, if $\delta < \min\{|c|/2, |c|^2\varepsilon/2\}$, then any $x$ satisfying $|x - c| < \delta$ must further satisfy

$$|x - c| < \delta < |c|^2\varepsilon/2 = (|c|/2)|c|\varepsilon < |x||c|\varepsilon,$$

which is the desired upper bound in (4.1). Thus, letting

$$\delta = 2^{-1} \min\{|c|/2, |c|^2\varepsilon/2\},$$

we see that whenever $x \in (c - \delta, c + \delta) \cap (\mathbb{R} \setminus \{0\})$, we have that $|f(x) - f(c)| < \varepsilon$. Hence, $f$ is continuous at $c$. □

**Lemma 4.6.** *Let $E \subseteq \mathbb{R}$ be a non-empty set, let $c \in E$, let $\alpha \in \mathbb{R}$, let $f : E \to \mathbb{R}$ be continuous at $c$ and let $f(c) > \alpha$. Then there exists some $\delta > 0$ such that $f(x) > \alpha$ for all $x \in (c-\delta, c+\delta) \cap E$.*

*Proof.* Since $f(c) > \alpha$, let $\varepsilon = (f(c) - \alpha)/2 > 0$. Since $f$ is continuous at $c$, there exists some $\delta > 0$ such that whenever $x \in (c-\delta, c+\delta) \cap E$, then $|f(x) - f(c)| < \varepsilon$. We can now deduce that

$$f(x) = f(c) + (f(x) - f(c)) \geqslant f(c) - |f(x) - f(c)| > f(c) - \varepsilon = (f(c) + \alpha)/2 > \alpha. \quad \square$$

A similar argument showed that if $f(c) < \alpha$ and $f$ is continuous at $c$, then there exists some $\delta > 0$ such that for all $x \in (c-\delta, c+\delta) \cap E$, we have $f(x) < \alpha$.

4.2. **Sequential continuity.** There is a (potentially) easier–to–apply notion of continuity.

**Definition 4.7.** A function $f : E \to \mathbb{R}$ is *sequentially continuous* at some point $c \in E$ if the following holds. Let $(x_n)_n$ be a sequence such that $x_n \in E$ for all $n \in \mathbb{N}$ and $x_n \to c$ as $n \to \infty$. Then $f(x_n) \to f(c)$ as $n \to \infty$.

**Lemma 4.8.** *If a function $f : E \to \mathbb{R}$ is continuous at $c \in E$, then it is sequentially continuous at $c$.*

*Proof.* Let $(x_n)_n$ be a sequence such that $x_n \in E$ for all $n \in \mathbb{N}$ and $x_n \to c$ as $n \to \infty$. Let $\varepsilon > 0$ be given. We need to find some $N \in \mathbb{N}$, such that whenever $n \geqslant N$, we have $|f(x_n) - f(c)| < \varepsilon$.

Since $f$ is continuous at $c$, for this choice of $\varepsilon$, there exists some $\delta > 0$ such that whenever $x \in E \cap (c-\delta, c+\delta)$, one has $|f(x) - f(c)| < \varepsilon$. Moreover, since $x_n \to c$, for this choice of $\delta > 0$, there exists some $N \in \mathbb{N}$ such that whenever $n \geqslant N$, one has $|x_n - c| < \delta$. This combines with the preceding statement to tell us that whenever $n \geqslant N$, one has $x_n \in E \cap (c-\delta, c+\delta)$, and so, $|f(x_n) - f(c)| < \varepsilon$. $\square$

It turns out that the above result is in fact an equivalence. We will prove this via contradiction, and so, we first need to understand what it means for a function to NOT be continuous at some point.

**Lemma 4.9.** *Let $f : E \to \mathbb{R}$ be a function and let $c \in E$. If $f$ is* discontinuous *at $c \in E$, then there exists some $\varepsilon > 0$ such that for any choice of $\delta > 0$, there must exist some $x \in (c-\delta, c+\delta) \cap E$ satisfying $|f(x) - f(c)| \geqslant \varepsilon$.*

*Proof.* This follows by negating the definition of continuity:

Recall that $f$ is continuous at $c$ if for every $\varepsilon$, the statement "$\exists \delta > 0$ such that $x \in E \cap (c-\delta, c+\delta) \Rightarrow |f(x) - f(c)| < \varepsilon$" holds true. If $f$ is discontinuous at $c$ then there must

exist some exceptional $\varepsilon > 0$ such that the above statement does not hold. Thus, fix this specific exceptional $\varepsilon > 0$.

Now, for the statement to NOT hold for this specific $\varepsilon$, we must have that for any $\delta > 0$, the mini-statement "$x \in E \cap (c-\delta, c+\delta) \Rightarrow |f(x) - f(c)| < \varepsilon$" does not hold. Indeed, if the mini-statement did hold for some $\delta > 0$, then we would get that the original statement holds for $\varepsilon$ and this specific choice of $\delta$.

Finally, fix any $\delta > 0$. We know that the the mini-statement for this $\delta$ is false. This means that there exists some $x \in E \cap (c-\delta, c+\delta)$ such that $|f(x) - f(c)| \geqslant \varepsilon$. Putting everything together gives the desired conclusion. □

We are now ready to prove that sequential continuity implies continuity.

**Lemma 4.10.** *Let $f : E \to \mathbb{R}$ be a function and let $c \in E$. If $f$ is sequentially continuous at $c$, then $f$ is continuous at $c$.*

*Proof.* We will proceed via contradiction, so suppose that $f$ is sequentially continuous at $c$, but $f$ is NOT continuous at $c$. Applying the definition of discontinuity, we see that there exists some $\varepsilon > 0$ such that for any $\delta > 0$, there exists some $x \in (c-\delta, c+\delta) \cap E$ such that $|f(x) - f(c)| \geqslant \varepsilon$. Fix this $\varepsilon > 0$.

Now for any $n \in \mathbb{N}$, we can apply the above with $\delta = 1/n$ to find some

$$x_n \in (c - 1/n, c + 1/n) \cap E$$

such that

$$|f(x_n) - f(c)| \geqslant \varepsilon. \tag{4.2}$$

Now consider the sequence $(x_n)_n$. Note that $x_n \in E$ for all $n \in \mathbb{N}$ and that for any $n \in \mathbb{N}$, one has

$$c - 1/n < x_n < c + 1/n.$$

By the sandwich rule, we see that $x_n \to c$. From the definition of sequential continuity, we get that $f(x_n) \to f(c)$. This means that for our previously fixed choice of $\varepsilon$, there must exist some $N \in \mathbb{N}$ such that whenever $n \geqslant N$, one has $|f(x_n) - f(c)| < \varepsilon$. This contradicts the fact that (4.2) holds for all $n \in \mathbb{N}$. □

**Example 4.11.** The function $f : \mathbb{R} \setminus \{4\} \to \mathbb{R}$ satisfying $f(x) = x^2 + 1/(x-4)$ for all $x \in \mathbb{R} \setminus \{4\}$ is continuous on $\mathbb{R} \setminus \{4\}$.

*Proof.* Indeed let $c \in \mathbb{R} \setminus \{4\}$, let $(x_n)_n$ be a sequence such that $x_n \in \mathbb{R} \setminus \{4\}$ for all $n \in \mathbb{N}$ and $x_n \to c$. Note that for any $n \in \mathbb{N}$, the number

$$f(x_n) = x_n^2 + 1/(x_n - 4)$$

is well-defined since $x_n \neq 4$ for all $n \in \mathbb{N}$. Since $c \neq 4$, we may apply algebra of limits to deduce that

$$f(x_n) = x_n^2 + 1/(x_n - 4) \to c^2 + 1/(c-4) = f(c).$$

This implies that $f$ is sequentially continuous at $c$, and hence, it is continuous at $c$. □

Given two functions $f, g : E \to \mathbb{R}$, we define the functions $fg : E \to \mathbb{R}$ and $f + g : E \to \mathbb{R}$ as

$$(f+g)(x) = f(x) + g(x) \quad \text{and} \quad (fg)(x) = f(x)g(x)$$

for all $x \in E$. If $g(x) \neq 0$ for all $x \in E$, we can also define $f/g : E \to \mathbb{R}$ as

$$(f/g)(x) = f(x)/g(x)$$

for all $x \in E$.

*Lecture 24*

**Lemma 4.12.** *Let $f, g : E \to \mathbb{R}$ be continuous at some point $c \in E$. Then the functions $f + g$, $fg$ are continuous at $c$. Moreover if $g(c) \neq 0$, then there exists some $\delta > 0$ such that the function $(f/g) : E \cap (c - \delta, c + \delta) \to \mathbb{R}$ is continuous at $c$.*

*Proof.* Let $(x_n)_n$ be a sequence such that $x_n \in E$ for all $n \in \mathbb{N}$ and $x_n \to c$. Since $f, g$ are continuous at $c$, we get that $f(x_n) \to f(c)$ and $g(x_n) \to g(c)$. By algebra of limits, we get that

$$(f+g)(x_n) = f(x_n) + g(x_n) \to f(c) + g(c) = (f+g)(c)$$

and

$$(fg)(x_n) = f(x_n)g(x_n) \to f(c)g(c) = (fg)(c).$$

Thus, $f + g$ and $fg$ are continuous at $c$.

As for $f/g$, note that since $g(c) \neq 0$, we may apply Lemma 4.6 (and the remark following Lemma 4.6) to deduce that there exists some $\delta > 0$ such that

$$g(x) \neq 0 \quad \text{for all} \quad x \in (c - \delta, c + \delta) \cap E. \tag{4.3}$$

Now let $(x_n)_n$ be some sequence such that $x_n \in (c - \delta, c + \delta) \cap E$ for all $n \in \mathbb{N}$ and $x_n \to c$. Since $f, g$ are continuous at $c$, we get that $f(x_n) \to f(c)$ and $g(x_n) \to g(c)$. Moreover, since $g(x_n) \neq 0$ for all $n \in \mathbb{N}$ (because of (4.3)) and $g(c) \neq 0$, we may apply algebra of limits to deduce that

$$(f/g)(x_n) = f(x_n)/g(x_n) \to f(c)/g(c) = (f/g)(c).$$

Thus the function $(f/g) : E \cap (c - \delta, c + \delta) \to \mathbb{R}$ is continuous at $c$. □

Lemma 4.12 allows us to construct a lot of continuous functions.

**Example 4.13.** (1) For any $d \in \mathbb{N}$, the function $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x) = x^d$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R}$.

(2) For any $d \in \mathbb{N}$ and any $a_0, a_1, \ldots, a_d \in \mathbb{R}$, the function $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x) = a_0 + a_1 x + \cdots + a_d x^d$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R}$.

(3) For any $d \in \mathbb{N}$ and any $a_{-d}, a_{-d+1}, \ldots, a_{d-1}, a_d \in \mathbb{R}$, the function $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ satisfying $f(x) = \sum_{i=-d}^{d} a_i x^i$ for all $x \in \mathbb{R}$ is continuous on $\mathbb{R} \setminus \{0\}$.

(4) Given polynomials $p(x)$ and $q(x)$, let $Z_q = \{x \in \mathbb{R} : q(x) = 0\}$ and define the *rational function* $p/q : \mathbb{R} \setminus Z_q \to \mathbb{R}$ as the function satisfying $(p/q)(x) = p(x)/q(x)$ for all $x \in \mathbb{R} \setminus Z_q$. Then $p/q$ is continuous on $\mathbb{R} \setminus Z_q$.

**Example* 4.14.** Given any $x \in \mathbb{R}$, we know from Example 3.25 that the series $\sum_{n=1}^{\infty} x^n/(n!)$ converges. Thus, let $f : \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x) = 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!}$$

for all $x \in \mathbb{R}$. Then $f$ is continuous on $\mathbb{R}$.

*Proof.* Let $c \in \mathbb{R}$, let $\varepsilon > 0$. Recall that by the ratio test, one has $(4|c| + 4)^n/n! \to 0$ as $n \to \infty$, and so, there exists some $N_1 \in \mathbb{N}$ be such that whenever $n \geqslant N_1$, one has

$$(4|c| + 4)^n/n! < \varepsilon/6.$$

Choose $N_2 = \max\{N_1, \lfloor 4|c| \rfloor + 4\}$.

Note that for any $k \geqslant N_2$ and any $x \in \mathbb{R}$ satisfying $|x| \leqslant 2|c|$, one has

$$\left| \sum_{n=N_2}^{k} \frac{x^n}{n!} \right| \leqslant \sum_{n=N_2}^{k} \frac{|x|^n}{n!} \leqslant \frac{(2|c|)^{N_2}}{N_2!} \sum_{n=N_2}^{k} \frac{(2|c|)^{n-N_2}}{(N_2+1)\ldots n}.$$

Our choice of $N_2$ implies that

$$\frac{(2|c|)^{N_2}}{N_2!} < \frac{\varepsilon}{6} \quad \text{and} \quad \frac{(2|c|)^{n-N_2}}{(N_2+1)\ldots n} \leqslant (1/2)^{n-N_2},$$

and so,

$$\left| \sum_{n=N_2}^{k} \frac{x^n}{n!} \right| < \frac{\varepsilon}{6} \sum_{n=N_2}^{k} (1/2)^{n-N_2} < \frac{\varepsilon}{3}.$$

Since limits preserve inequalities, we get that

$$-\frac{\varepsilon}{3} \leqslant f(x) - \sum_{n<N_2} \frac{x^n}{n!} \leqslant \frac{\varepsilon}{3} \tag{4.4}$$

for any $x \in \mathbb{R}$ with $|x| \leqslant 2|c|$.

We are now ready to finish our proof. Since $\sum_{n<N_2} x^n/n!$ is a polynomial and polynomials are continuous on $\mathbb{R}$, there exists some $\delta' > 0$ such that whenever $x \in (c-\delta, c+\delta)$, then

$$|\sum_{n<N_2} x^n/n! - \sum_{n<N_2} c^n/n!| < \varepsilon/3.$$

Choose $\delta = \min\{\delta', |c|/2\}$. Note that for any $x \in (c-\delta, c+\delta)$, one has $|x| < 2|c|$, and so, (4.4) holds for all such $x$. Moreover, noting the preceding inequality, we may apply triangle inequality to deduce that

$$\begin{aligned} |f(x) - f(c)| &\leqslant |f(x) - \sum_{n<N_2} x^n/n!| + |\sum_{n<N_2} x^n/n! - \sum_{n<N_2} c^n/n!| + |\sum_{n<N_2} c^n/n! - f(c)| \\ &< \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

□

*Remark** 4.15. A key ingredient in the above proof was (4.4), that is, the fact that given $\varepsilon > 0$, there existed an $N_2 \in \mathbb{N}$ such that $|f(x) - \sum_{n<N_2} x^n/(n!)| < \varepsilon$ for all $x \in [-2|c|, 2|c|]$. A priori there is no reason for this $N_2$ to be independent of our choice of $x$ in the set $[-2|c|, 2|c|]$. The fact that we could prove this implies that not only is the series $\sum_n x^n/n!$ convergent for all $x \in [-2|c|, 2|c|]$, it is *uniformly convergent*.

4.3. **Discontinuity.** Recall that $f : E \to \mathbb{R}$ is *discontinuous* at $c \in E$ if there exists some $\varepsilon > 0$ such that for any choice of $\delta > 0$, there must exist some $x \in (c-\delta, c+\delta) \cap E$ satisfying $|f(x) - f(c)| \geqslant \varepsilon$.

In fact, using sequential continuity, we can also prove that $f : E \to \mathbb{R}$ is *discontinuous* at $c \in E$ if there exists some sequence $(x_n)_n$ with $x_n \in E$ for all $n \in \mathbb{N}$ such that $x_n \to c$ but $(f(x_n))_n$ does not converge to $f(c)$.

**Example 4.16.** The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} -1 & x \leqslant 0 \\ 1 & x > 0 \end{cases}$$

is not continuous at $x = 0$.

*Proof.* Indeed, let $x_n = (-1)^n/n$. Note that $x_n \to 0$ but $f(x_n) = (-1)^n$ for all $n \in \mathbb{N}$ and this sequence does not converge. □

**Example 4.17.** The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \lfloor x \rfloor$ is discontinuous at any $c \in \mathbb{Z}$.

*Proof.* Indeed, let $c \in \mathbb{Z}$ and let $x_n = c - 1/n$ for all $n \in \mathbb{N}$. Then $x_n \to c$ but $f(x_n) = c - 1$ for all $n \in \mathbb{N}$ while $f(c) = c$. □

Alternatively, let $\varepsilon = 1/2$. Then for any $\delta > 0$, we may choose $x = c - \delta/2 \in (c - \delta, c + \delta)$, and note that $f(x) = c - 1$ and $|f(x) - f(c)| = 1 > 1/2$. Thus $f$ is discontinuous at $c$.

*Remark* 4.18. Let $E = \mathbb{Z}$ and let $g : E \to \mathbb{R}$ be a function. Then $g$ is continuous on $E$. Indeed for any $c \in \mathbb{Z}$ and $\varepsilon > 0$, we may set $\delta = 1/10$ and note that if $x \in (c - \delta, c + \delta) \cap E$, then $x = c$, in which case $|g(x) - g(c)| = 0 < \varepsilon$.

Here are some more examples.

**Example 4.19.** Let $g, h : \mathbb{R} \to \mathbb{R}$ be continuous functions. Let

$$f(x) = \begin{cases} g(x) & x \in \mathbb{Q} \\ h(x) & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Then $f$ is continuous at some $c \in \mathbb{R}$ if and only if $g(c) = h(c)$.

*Proof.* Suppose $c \in \mathbb{R}$ satisfies $g(c) \neq h(c)$. Let $(x_n)_n, (y_n)_n$ be sequences such that $x_n \in \mathbb{Q}$ and $y_n \in \mathbb{R} \setminus \mathbb{Q}$ for all $n \in \mathbb{N}$ and $x_n \to c$ and $y_n \to c$. Since $g, h$ are continuous, we get that $f(x_n) = g(x_n) \to g(c)$ while $f(y_n) = h(y_n) \to h(c)$. If $c \in \mathbb{Q}$, the fact that $y_n \to c$ but $f(y_n) \to h(c) \neq f(c)$ gives us a contradiction. One can similarly handle the case when $c \in \mathbb{R} \setminus \mathbb{Q}$.

Now suppose $c \in \mathbb{R}$ satisfies $g(c) = h(c)$. Let $\varepsilon > 0$ be given. There exist $\delta_1, \delta_2$ such that whenever $|x - c| < \delta_1$ then $|g(x) - g(c)| < \varepsilon$, and whenever $|x - c| < \delta_2$, then $|h(x) - h(c)| < \varepsilon$. Choose $\delta = \min\{\delta_1, \delta_2\}$ and note that if $x \in (c - \delta, c + \delta) \cap \mathbb{Q}$, then $|f(x) - f(c)| = |g(x) - g(c)| < \varepsilon$. The case when $x \in (c - \delta, c + \delta) \cap (\mathbb{R} \setminus \mathbb{Q})$ can be dealt with similarly. □

**Example 4.20.** Let $f : (0, 1) \to \mathbb{R}$ be a function defined as

$$f(x) = \begin{cases} 1/q & x = p/q \text{ for some co-prime } p, q \in \mathbb{N} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

The function $f$ is discontinuous at all rational numbers and $f$ is continuous at all irrational numbers.

*Proof.* Let $c = p/q$ for some co-prime $p, q \in \mathbb{N}$. Let $(x_n)_n$ be a sequence with $x_n \in \mathbb{R} \setminus \mathbb{Q}$ for all $n \in \mathbb{N}$ and $x_n \to c$. Then $f(x_n) = 0$ for all $n \in \mathbb{N}$ while $f(c) = 1/q \neq 0$. Thus, $(f(x_n))_n$ does not converge to $f(c)$ and so, $f$ is discontinuous at $c$.

Let $c \in (0,1) \setminus \mathbb{Q}$ and let $\varepsilon > 0$. We want to find some $\delta > 0$ such that for all $x \in (c-\delta, c+\delta) \cap (0,1)$, we have $|f(x) - f(c)| = |f(x)| < \varepsilon$. If such an $x$ is irrational, then $f(x) = 0$, and so $|f(x)| < \varepsilon$ for ANY choice of $\delta > 0$. Thus our choice of $\delta$ is going to be determined by the case when $x \in \mathbb{Q}$. Writing $x = p/q$, for some co-prime integers $p, q \geqslant 1$, we get that $|f(x)| = 1/q$. Thus, we need to find a $\delta > 0$ such that whenever $p/q \in (c-\delta, c+\delta) \cap (0,1)$ for co-prime $p, q \geqslant 1$, one has $q > 1/\varepsilon$. Since $c$ itself is irrational, one way to obtain this $\delta$ is by letting

$$S = \{p/q : 1 \leqslant p \leqslant q \leqslant 1/\varepsilon + 1\}$$

and setting

$$\delta = \min_{s \in S} |c - s|/2,$$

see (0.1) for the "$\min_{s \in S}$" notation. Note that this is well-defined since $S$ is finite: there are at most $(1/\varepsilon+1)$ choices of $q$ and at most $(1/\varepsilon+1)$ choices of $p$ satisfying $1 \leqslant p \leqslant q \leqslant 1/\varepsilon+1$.

Firstly, since $c \in (0,1) \setminus \mathbb{Q}$, we see that $\delta > 0$. Moreover, if $x \in \mathbb{Q} \cap (c-\delta, c+\delta)$, then the definition of $\delta$ implies that $x \notin S$; indeed, $|x - c| < \delta \leqslant 2|s - c|$ for all $s \in S$. This means that upon writing such an $x$ as $x = p/q$ with co-prime $p, q \geqslant 1$, we must have $q > 1/\varepsilon + 1$. This in turn implies that if $x \in \mathbb{Q} \cap (c-\delta, c+\delta)$, then $|f(x)| = 1/q < \varepsilon$, and so, we are done. □

*Lecture 25*

4.4. **Continuous functions on intervals.** We will now see that continuous functions exhibit nice properties on closed intervals.

**Intermediate Value theorem.** *Let $a < b$ be real numbers and let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. If $f(a) < f(b)$, then for all $g \in (f(a), f(b))$, there exists some $c \in (a,b)$ such that $f(c) = g$.*

*Similarly, if $f(a) > f(b)$, then for all $g \in (f(b), f(a))$, there exists some $c \in (a,b)$ such that $f(c) = g$.*

*Remark* 4.21. Note that this immediately implies that there exists some $c \in \mathbb{R}$ such that $c^2 = 2$. Indeed the function $f(x) = x^2 - 2$ is continuous on $[1,2]$ and $f(1) = -1 < 0 < 2 = f(2)$, and so, there must exist $c \in (1,2)$ such that $c^2 - 2 = 0$. This, in turn, suggests that the proof of the Intermediate value theorem must use the completeness of real numbers.

*Proof of Intermediate value theorem.* We will just prove the result when $f(a) < f(b)$, the other case will follow similarly. Given $g \in (f(a), f(b))$, let

$$S = \{x \in [a, b] : f(x) < g\}.$$

Note that $S$ is non-empty, indeed $a \in S$. Moreover, $S$ is bounded above, since for all $x \in S$, one has $s \leqslant b$. Thus by the least upper bound axiom, $\sup S$ exists. Let $c = \sup S$. We quickly observe that $c \in [a, b]$: $c \geqslant a$ since $a \in A$, and $c \leqslant b$ since $b$ is an upper bound for $S$.

We now claim that $f(c) = g$. In particular, it suffices to show that the statements $f(c) > g$ and $f(c) < g$ cannot hold.

We first consider the former case, and so, assume for contradiction that $f(c) > g$. Since $f(a) < g$, we get that $c \in (a, b]$. Let $\varepsilon = (f(c) - g)/2 > 0$ and note that since $f$ is continuous on $[a, b]$ and $c \in [a, b]$, there exists a $\delta' > 0$ such that for all $x \in (c - \delta', c] \cap [a, b]$, one has $|f(x) - f(c)| < \varepsilon$. Choosing $\delta = 2^{-1} \min\{\delta', c - a\}$, we see that $(c - \delta, c] \subseteq (a, b]$ and for all $x \in (c - \delta, c]$, one has

$$f(x) = f(c) + (f(x) - f(c)) \geqslant f(c) - |f(x) - f(c)| > f(c) - \varepsilon > g.$$

On the other hand, since $c = \sup S$, there exists some $s \in S$ such that $s \in (c - \delta, c]$, but the above inequality would imply that $g > f(s) > g$, delivering the desired contradiction.

We now consider the case when $f(c) < g$. In this case, since $f(b) > g$, we get that $c \in [a, b)$. Now let $\varepsilon = (g - f(c))/2$, and as before, we may find some $\delta > 0$ such that $[c, c + \delta) \subseteq [a, b)$ and for all $x \in [c, c + \delta)$, one has $|f(x) - f(c)| < \varepsilon$. The latter condition implies that

$$f(x) = f(c) + f(x) - f(c) < f(c) + |f(x) - f(c)| < f(c) + \varepsilon < g.$$

This means that $c + \delta/2 \in [a, b]$ and $f(c + \delta/2) < g$, and so, $c + \delta/2 \in S$, contradicting the fact that $c$ is an upper bound for $S$. □

Intermediate value theorem can be quite useful for exhibiting real solutions to certain equations.

**Example 4.22.** There exists some $c \in \mathbb{R}$ such that $c^3 + 10 = 0$.

*Proof.* Indeed, let $f(x) = x^3 + 10$ and note that $f$ is continuous on $[-10, 0]$. Moreover, $f(-10) < 0 < f(0)$ and so, there exists some $c \in (-10, 0)$ such that $f(c) = 0$. □

**Example 4.23.** There exists some $c \in \mathbb{R}$ such that $ce^c = 1$.

*Proof.* In a later module, you will learn that the function $f(x) = e^x$ is continuous on $\mathbb{R}$. With this information in hand, define the function $g(x) = xe^x - 1$ for all $x \in \mathbb{R}$. By Lemma 4.12, we can prove that $g$ is continuous on $[0, 1]$. Moreover, note that $g(0) = -1 < 0 < e - 1 = g(1)$,

and so, Intermediate value theorem implies that there exists some $c \in (0,1)$ such that $ce^c = 1$. $\square$

*Remark* 4.24. It is quite important that $f$ is continuous on the entirety of $[a,b]$ in the hypothesis of Intermediate value theorem. For example, consider the function $f(x) = x$ whenever $x > 0$ and $f(x) = -1$ whenever $x \leqslant 0$. Note that $f$ is continuous on $(0,1]$ and $f(0) = -1 < -1/2 < 1 = f(1)$, but there is no $c \in \mathbb{R}$ such that $f(c) = -1/2$.

*Lecture 26*

**Definition 4.25.** Let $E \subseteq \mathbb{R}$ be a non-empty set. A function $f : E \to \mathbb{R}$ is *bounded above on* $E$ if there exists $M \in \mathbb{R}$ such that $f(x) \leqslant M$ for all $x \in E$. Similarly, $f$ is *bounded below on* $E$ if there exists $m \in \mathbb{R}$ such that $f(x) \geqslant m$ for all $x \in E$. Finally, $f$ is *bounded on* $E$ if it is bounded above and bounded below on $E$.

Given a non-empty set $E \subseteq \mathbb{R}$ and bounded function $f : E \to \mathbb{R}$, we see that the set $f(E) = \{f(x) : x \in E\}$ is non-empty and bounded, and so $\sup f(E)$ and $\inf f(E)$ exist. We write

$$\sup_{x \in E} f(x) = \sup f(E) \quad \text{and} \quad \inf_{x \in E} f(x) = \inf f(E).$$

**Extreme Value Theorem.** *Let $a < b$ be real numbers, let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then $f$ is bounded and attains its bounds, that is, there exist $u, v \in [a,b]$ such that*

$$f(u) = \inf_{x \in [a,b]} f(x) \quad \textit{and} \quad f(v) = \sup_{x \in [a,b]} f(x).$$

The above conclusion can be further rewritten as: there exist $u, v \in [a,b]$ such that for all $x \in [a,b]$, one has $f(u) \leqslant f(x) \leqslant f(v)$.

*Remark* 4.26. It is quite important that we have a closed interval in the hypothesis of the Extreme value theorem. Indeed consider the set $E = (0,1)$ and $f(x) = x$. $f$ does not attain its infimum or supremum on $E$. Similarly take $E = (0,1)$ and $f(x) = 1/x$, and note that $f$ is continuous on $E$ but $f$ is not bounded above on $E$.

*Proof of Extreme value theorem.* It suffices to prove that there exists some $v \in [a,b]$ such that $f(x) \leqslant f(v)$ for all $x \in [a,b]$. Indeed, one can get the lower bound result by just replacing $f$ by $-f$.

We will first prove that $f([a,b])$ is bounded above, and we do so via contradiction. Thus, suppose that $f([a,b])$ is NOT bounded above, and so, for every $n \in \mathbb{N}$, there must exist some $x_n \in [a,b]$ such that $f(x_n) > n$. Since the sequence $(x_n)_n$ is bounded, we may apply

Bolzano–Weierstrass Theorem to obtain a subsequence $(x_{n_j})_j$ such that $x_{n_j} \to l$ as $n_j \to \infty$ for some $l \in \mathbb{R}$. Note that since $a \leqslant x_{n_j} \leqslant b$ for all $j \in \mathbb{N}$, we get that $a \leqslant l \leqslant b$. Now, since $l \in [a, b]$, we get that $f(l)$ is some real number and that $f$ is continuous at $l$. Moreover, since $x_{n_j} \to l$, we must have $f(x_{n_j}) \to f(l)$. On the other hand, by definition, we have that $f(x_{n_j}) > n_j \geqslant j$ for every $j \in \mathbb{N}$ and so $f(x_{n_j}) \to \infty$. This contradicts the fact that $f(x_{n_j}) \to f(l)$. Hence we have proved that $f([a, b])$ is bounded above.

We will now prove that there exists some $v \in [a, b]$ such that $f(x) \leqslant f(v)$ for all $x \in [a, b]$. In this endeavour, note that the set $f(E)$ is non-empty and bounded above, so it must have a supremum. Let $M = \sup f(E)$. By definition of supremum, for any $n \in \mathbb{N}$, there exists some $y_n \in [a, b]$ such that $M - 1/n < f(y_n) \leqslant M$. Note that the sequence $(y_n)$ is bounded and so, by Bolzano–Weierstrass Theorem, contains a convergent subsequence $(y_{m_j})_j$. As before, this means that there exists some $v \in [a, b]$ such that $y_{m_j} \to v$. Since $v \in [a, b]$, we get that $f(v)$ is a real number and that $f$ is continuous at $v$, and so, $f(y_{m_j}) \to f(v)$. On the other hand, note that $m_j \to \infty$ as $j \to \infty$ and

$$M - 1/m_j < f(y_{m_j}) \leqslant M$$

for all $j \in \mathbb{N}$. By the sandwich rule, we get that $f(v) = \lim_{j\to\infty} f(y_{m_j}) = M$. This is precisely the desired conclusion. □

*Lecture 27*

4.5. **Continuous functions preserve intervals.** The Extreme value theorem maybe combined with the Intermediate value theorem to see that for any real numbers $a < b$ and for any function $f : [a, b] \to \mathbb{R}$ which is continuous on $[a, b]$, the set $f([a, b])$ is itself a closed interval of the form $[f(u), f(v)]$ where $u, v \in [a, b]$. We will prove this along with some related statements.

**Definition 4.27.** Given a non-empty set $E \subseteq \mathbb{R}$, we say that $E$ is an *interval* if for all $x, y \in E$ with $x \leqslant y$, we also have $[x, y] \subseteq E$.

**Example 4.28.** The sets $(-\infty, 100), (1, 2), [-5, -2), [0, \infty), \{34\}, \mathbb{R}, [10, 12]$ are all examples of intervals.

Using Intermediate value theorem, we can show that continuous function preserve intervals.

**Lemma 4.29.** *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be a continuous function on $I$. Then $f(I)$ is also an interval.*

*Proof.* Let $u \leqslant v$ be real numbers such that $u, v \in f(I)$. We want to prove that $[u, v] \subseteq f(I)$.

Note that if $u = v$, then we trivially have $[u, v]$ $\{u\} \subseteq f(I)$, and so, we may assume that $u < v$. By definition of $f(I)$, this means that there exist $x, y \in I$ such that $f(x) = u$ and $f(y) = v$. Since $u \neq v$, we must have $x \neq y$.

We suppose that $x < y$; the case when $x > y$ can be dealt with in a similar fashion. Since $I$ is an interval, we get that $[x, y] \subseteq I$. Since $f$ is continuous on $I$, we get that $f$ is continuous on $[x, y]$. We may now apply the Intermediate value theorem to deduce that given any $u < g < v$, there must exist some $x < c < y$ such that $f(c) = g$. This means that $[u, v] \subseteq f(I)$, and so, we are done. □

**Lemma 4.30.** *Let $a \leqslant b$ be real numbers, let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then there exist $c, d \in \mathbb{R}$ such that $f([a, b]) = [c, d]$.*

*Proof.* The case when $a = b$ is trivial and so we may assume that $a < b$. By Extreme value theorem, we know that there exist $u, v \in [a, b]$ such that for all $x \in [a, b]$, one has $f(u) \leqslant f(x) \leqslant f(v)$, that is, $f(I) \subseteq [f(u), f(v)]$. Our aim now is to show that $f(I) = [f(u), f(v)]$, and in particular, it suffices to show that $[f(u), f(v)] \subseteq f(I)$.

The case when $u = v$ is trivial, since this implies that $f(x) = f(u)$ for all $x \in [a, b]$ in which case we are done. We may assume that $u > v$; the case when $u < v$ may be dealt with mutatis mutandis. Since $f$ is continuous on $[a, b]$, we get that $f$ is continuous on $[v, u]$. As in the proof of Lemma 4.29, we may now apply intermediate value theorem to deduce that for all $f(u) < g < f(v)$, there exists some $c \in [v, u]$ such that $f(c) = g$. In particular, $[f(u), f(v)] \subseteq f(I)$. This concludes the proof of Lemma 4.30 □

4.6. **Real solutions of polynomial equations.** Intermediate value theorem and Extreme value theorem can be used to prove that odd–degree polynomial equations always have a real solution and that even–degree polynomials with a positive leading coefficients achieve a global minimum over reals.

**Proposition 4.31.** *Let $k \in \mathbb{N}$, let $a_0, \ldots, a_{2k-1}$ be real numbers such that $a_{2k-1} \neq 0$. Then there exists some $c \in \mathbb{R}$ such that*

$$a_{2k-1}c^{2k-1} + a_{2k-2}c^{2k-2} + \cdots + a_1 c + a_0 = 0.$$

*Proof.* Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined as

$$f(x) = a_{2k-1}x^{2k-1} + a_{2k-2}x^{2k-2} + \cdots + a_1 x + a_0$$

for all $x \in \mathbb{R}$. Since $f$ is a polynomial, we get that $f$ is continuous on $\mathbb{R}$. We will assume that $a_{2k-1} > 0$; the case when $a_{2k-1} < 0$ can be analysed similarly.

We will first show that there exists some $M_1 \in \mathbb{N}$ such that $f(M_1) \geqslant 10$. In order to prove this, we first use Algebra of limits to see that

$$\lim_{n\to\infty} \frac{f(n)}{n^{2k-1}} = a_{2k-1} > 0,$$

whence, there exists $N_1 \in \mathbb{N}$ such that $f(n) \geqslant a_{2k-1}n^{2k-1}/2$ for all $n \geqslant N_1$. Next, since $n^{2k-1} \to \infty$ as $n \to \infty$, we get that there exists $N_2 \in \mathbb{N}$ such that $n^{2k-1} > 20/a_{2k-1}$ for any $n \geqslant N_2$. Letting $M_1 = \max\{N_1, N_2\}$, we see that

$$f(M_1) \geqslant a_{2k-1}M_1^{2k-1}/2 \geqslant 10.$$

Similarly, noting the fact that

$$\lim_{n\to\infty} f(-n)/n^{2k-1} = -a_{2k-1} < 0,$$

we can prove that there exists some $M_2 \in \mathbb{N}$ such that

$$f(M_2) \leqslant -10.$$

Now, since $f$ is continuous on $\mathbb{R}$, we get that $f$ is continuous on $[-M_2, M_1]$. Moreover, as $f(-M_2) \leqslant -10 < 0 < 10 \leqslant f(M_1)$, we may apply the Intermediate value theorem to find some $c \in (-M_2, M_1)$ such that $f(c) = 0$. □

*Lecture 28*

**Lemma 4.32.** *Let $k \in \mathbb{N}$, let $a_0, \ldots, a_{2k}$ be real numbers such that $a_{2k} > 0$. Let $f : \mathbb{R} \to \mathbb{R}$ be the function satisfying*

$$f(x) = a_{2k}x^{2k} + a_{2k-1}x^{2k-1} + \cdots + a_1x + a_0$$

*for all $x \in \mathbb{R}$. Let $R > 0$. Then there exists some $N \in \mathbb{N}$ such that whenever $x \in \mathbb{R}$ satisfies $|x| \geqslant N$, we have $f(x) \geqslant R$.*

*Proof.* Since $a_{2k} > 0$ and $x^{2k} \geqslant 0$ for all $x \in \mathbb{R}$, we may apply triangle inequality to deduce that for all $x \in \mathbb{R}$, one has

$$\begin{aligned} f(x) - a_{2k}x^{2k} &\geqslant -|f(x) - a_{2k}x^{2k}| = -|a_{2k-1}x^{2k-1} + \cdots + a_1x + a_0| \\ &\geqslant -|a_{2k-1}||x|^{2k-1} - \cdots - |a_1||x| - |a_0|. \end{aligned}$$

In fact, for all $x \in \mathbb{R}$ with $|x| \geqslant 1$, the above inequality implies that

$$f(x) \geqslant a_{2k}x^{2k} - (|a_{2k-1}| + \cdots + |a_0|)|x|^{2k-1}. \tag{4.5}$$

Letting $A = |a_{2k-1}| + \cdots + |a_0| \geqslant 0$, we define

$$P(x) = a_{2k}x^{2k} - A|x|^{2k-1}$$

for all $x \in \mathbb{R}$. Using algebra of limits, we can show that $P(n)/n^{2k} \to a_{2k} > 0$. Combining this with the fact that $n^{2k} \to \infty$ as $n \to \infty$, we can deduce that there exists some $N \in \mathbb{N}$ such that $P(N) > R$. Moreover, for any $x \in \mathbb{R}$ with $|x| \geqslant N$, note that

$$x^{2k} \geqslant N^{2k} \quad \text{and} \quad a_{2k} - A/|x| \geqslant a_{2k} - A/N,$$

and so,

$$P(x) = x^{2k}(a_{2k} - A/|x|) \geqslant N^{2k}(a_{2k} - A/|N|) = P(N) > R.$$

Noting (4.5), we see that whenever $x \in \mathbb{R}$ satisfies $|x| \geqslant N$, one has $f(x) \geqslant P(x) > R$. □

We are now ready to prove our main result about even degree polynomials with positive leading coefficients.

**Proposition 4.33.** *Let $k \in \mathbb{N}$, let $a_0, \ldots, a_{2k}$ be real numbers such that $a_{2k} > 0$. Let $f : \mathbb{R} \to \mathbb{R}$ be the function satisfying*

$$f(x) = a_{2k}x^{2k} + a_{2k-1}x^{2k-1} + \cdots + a_1x + a_0$$

*for all $x \in \mathbb{R}$. Then there exists some $u \in \mathbb{R}$ such that for all $x \in \mathbb{R}$, one has $f(x) \geqslant f(u)$.*

*If $f(u) > 0$, then $f$ has no real roots. If $f(u) = 0$, then $f$ has at least one real root. If $f(u) < 0$, then $f$ has at least two real roots.*

*Proof.* We apply Lemma 4.32 to obtain some $N \in \mathbb{N}$ such that whenever $x \in \mathbb{R}$ satisfies $|x| \geqslant N$, we have $f(x) \geqslant 2|a_0| + 1 > a_0 = f(0)$. Next, since $f$ is a polynomial, we get that $f$ is continuous on $[-N, N]$, and so, Extreme value theorem tells us that there exists some $u \in [-N, N]$ such that $f(x) \geqslant f(u)$ for all $x \in [-N, N]$. In particular, we have $f(0) \geqslant f(u)$. Thus, for any $x \in \mathbb{R}$ with $|x| \geqslant N$, one has $f(x) > f(0) \geqslant f(u)$. Hence, we have shown that for any $x \in \mathbb{R}$, one has $f(x) \geqslant f(u)$.

Now, if $f(u) > 0$, then for all $x \in \mathbb{R}$, we must have $f(x) \geqslant f(u) > 0$, which means that $f(x) \neq 0$ for all $x \in \mathbb{R}$. Next, if $f(u) = 0$, then indeed $f$ has at least one real root. Finally, we consider the case when $f(u) < 0$. Recall that $u \in [-N, N]$ and $f(2N) \geqslant 2|a_0|+1 > 1$ and $f(-2N) \geqslant 2|a_0| + 1 > 1$. Thus, we may apply Intermediate value theorem to deduce that there exists some $c_1 \in (-2N, u)$ such that $f(c_1) = 0$. Similarly, Intermediate value theorem tells us that there exists some $c_2 \in (u, 2N)$ such that $f(c_2) = 0$. Thus, $f$ has at least two real roots. □

4.7. **Continuous Inverses.** We are now interested in investigating when can we "invert" continuous functions and still obtain continuous functions.

**Definition 4.34.** Given function $f : E \to \mathbb{R}$, we say that $f$ is *strictly increasing* if whenever $x, y \in E$ satisfy $x < y$, then $f(x) < f(y)$. We say that $f$ is *strictly decreasing* if whenever $x, y \in E$ satisfy $x < y$, then $f(x) > f(y)$. If $f$ is either strictly decreasing or strictly increasing, we say that $f$ is *strictly monotonic*.

**Definition 4.35.** Given non-empty sets $A, B \subseteq \mathbb{R}$, we say that $f : A \to B$ is *injective* (or equivalently *one-to-one*) if whenever $x, y \in A$ satisfy $f(x) = f(y)$, then $x = y$. We say that $f$ is *surjective* (or equivalently *onto*) if for all $y \in B$ there exists $x \in A$ such that $f(x) = y$. We say that $f$ is a *bijective* if $f$ is both injective and surjective.

**Definition 4.36.** Let $f : E \to \mathbb{R}$ be injective. Observe that for any $y \in f(E)$, there exists a unique $x_y \in E$ such that $f(x_y) = y$. Thus, we define the function $f^{-1} : f(E) \to \mathbb{R}$ as $f^{-1}(y) = x_y$ for all $y \in f(E)$.

**Lemma 4.37.** *Let $f : E \to \mathbb{R}$ be a strictly increasing function. Then $f : E \to f(E)$ is injective and $f^{-1} : f(E) \to E$ is strictly increasing.*

*Proof.* Suppose $x, y \in E$ satisfy $f(x) = f(y)$. Since $f$ is strictly increasing, we get that if $x > y$ then $f(x) > f(y)$, and if $x < y$ then $f(x) < f(y)$. Thus we must have $x = y$. This implies that $f$ is injective.

We will now prove that $f^{-1} : f(E) \to E$ is strictly increasing, and we prove this by contradiction. Thus suppose $y, z \in E$ satisfy $y < z$ and $f^{-1}(y) \geqslant f^{-1}(z)$. Recall that $f^{-1}(y) = x_y$ and $f^{-1}(z) = x_z$, and so, we get that $x_y \geqslant x_z$. Since $f$ is strictly increasing, this means that $f(x_y) \geqslant f(x_z)$, and so, we get that $y \geqslant z$, which contradicts our assumption that $y < z$. □

**Theorem 4.38** (Inverse function theorem for continuous functions)**.** *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous and strictly increasing. Then $J = f(I)$ is an interval and $f^{-1} : J \to I$ is continuous and strictly increasing.*

*A similar statement holds if $f$ is continuous and strictly decreasing, in which case we will get that $f^{-1}$ is continuous and strictly decreasing.*

*Proof.* We will prove our result for the strictly increasing case, the strictly decreasing case can be derived by replacing $f$ with $-f$.

We have already proven that $J = f(I)$ must be an interval, see Lemma 4.29. Moreover, Lemma 4.37 implies that $f$ is injective and $f^{-1} : J \to I$ is strictly increasing. Thus, it suffices to prove that $f^{-1} : J \to I$ is continuous.

Note that since $f$ is strictly increasing, $I$ is a singleton if and only if $J$ is a singleton. Moreover, if $J$ is a singleton, then one can swiftly check the definition of continuity to see that $f^{-1} : J \to I$ is continuous. Thus, we may assume that neither $I$ nor $J$ is a singleton.

Given $c \in J$, we want to show that $f^{-1}$ is continuous at $c$. Suppose first that $x_c = f^{-1}(c)$ is not an end point of $I$, that is, there exists some $\eta > 0$ such that $(x_c - \eta, x_c + \eta) \subseteq I$. Given $\varepsilon > 0$, we define $\varepsilon' = \min\{\varepsilon, \eta\}$. Since $f$ is strictly increasing, we get that

$$f(x_c - \varepsilon') < f(x_c) = c < f(x_c + \varepsilon').$$

Let

$$\delta = \min\{f(x_c) - f(x_c - \varepsilon'), f(x_c + \varepsilon') - f(x_c)\} > 0.$$

This implies that

$$c + \delta = f(x_c) + \delta \leqslant f(x_c + \varepsilon') \quad \text{and} \quad c - \delta = f(x_c) - \delta \geqslant f(x_c - \varepsilon'). \tag{4.6}$$

Now, note that if $y \in (c-\delta, c+\delta) \cap J$, then either $y = c$ or $c < y < c+\delta$ or $c-\delta < y < c$. In the first case, we get that $f^{-1}(y) - f^{-1}(c) = 0$. In the second case, that is when $c < y < c+\delta$, since $f^{-1}$ is strictly increasing, we get that

$$0 < f^{-1}(y) - f^{-1}(c) < f^{-1}(c + \delta) - f^{-1}(c) \leqslant f^{-1}(f(x_c + \varepsilon')) - x_c = \varepsilon' < \varepsilon,$$

where the penultimate inequality follows from (4.6). Similarly, in the third case, that is, when $c - \delta < y < c$, since $f^{-1}$ is strictly increasing, we get that

$$0 < f^{-1}(c) - f^{-1}(y) < f^{-1}(c) - f^{-1}(c - \delta) \leqslant x_c - f^{-1}(f(x_c - \varepsilon')) = \varepsilon' < \varepsilon,$$

where, as before, the penultimate inequality follows from (4.6).

The case when $x_c = f^{-1}(c)$ is an end point of $I$ can be analysed similarly. This concludes the proof of Theorem 4.38. □

## 5. Construction of Real numbers (Non-examinable)

The following section of the notes is completely non-examinable. Moreover, we might use some notation in this section which is not standard.

For the purposes of this section [1], we will only assume the existence of rational numbers and the fact that we understand basic arithmetic operations as well as the notion of order on rational numbers. Moreover, in this section, we will simply provide the basic ideas used in the construction of real numbers and not prove anything rigorously[2].

**Definition 5.1.** We say that a sequence $(a_n)_n$ with $a_n \in \mathbb{Q}$ for all $n \in \mathbb{N}$ is $\mathbb{Q}$–*Cauchy* if for any $\varepsilon \in \mathbb{Q}$ with $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that whenever $n, m \geqslant N$, we have $|a_n - a_m| < \varepsilon$.

We say that $(a_n)_n$ is $\mathbb{Q}$–*bounded above* if there exists some $M \in \mathbb{Q}$ such that $a_n \leqslant M$ for all $n \in \mathbb{N}$. Similarly, we can define $\mathbb{Q}$–*bounded below* and $\mathbb{Q}$–*bounded*.

Two sequences $(a_n), (b_n)$ with $a_n, b_n \in \mathbb{Q}$ for all $n \in \mathbb{N}$ are *equivalent* if for any $\varepsilon \in \mathbb{Q}$ with $\varepsilon > 0$, there exists some $N \in \mathbb{N}$ such that $|a_n - b_n| < \varepsilon$ holds for all $n \geqslant N$. We may write this as $(a_n) \sim (b_n)$.

One can swiftly prove the following familiar properties of $\mathbb{Q}$–Cauchy sequences.

**Lemma 5.2.** *If $(a_n)$ is $\mathbb{Q}$–Cauchy, then $(a_n)$ is $\mathbb{Q}$–bounded.*

**Lemma 5.3.** *Suppose $(a_n) \sim (b_n)$. Then $(a_n)$ is $\mathbb{Q}$–Cauchy if and only if $(b_n)$ is $\mathbb{Q}$–Cauchy.*

**Lemma 5.4.** *Suppose $(a_n), (b_n)$ are $\mathbb{Q}$–Cauchy. Then $(a_n + b_n)$ and $(a_n b_n)$ are $\mathbb{Q}$–Cauchy. Moreover, for any $\lambda \in \mathbb{Q}$, the sequence $(\lambda a_n)$ is $\mathbb{Q}$–Cauchy.*

**Definition 5.5.** Given any $\mathbb{Q}$–Cauchy sequence $(a_n)$, we define $\mathrm{LIM}(a_n)$ to be an object such that whenever $(a_n) \sim (b_n)$ then $\mathrm{LIM}(a_n) = \mathrm{LIM}(b_n)$.

Let $(a_n), (b_n)$ be $\mathbb{Q}$–Cauchy and let $x = \mathrm{LIM}(a_n)$ and $y = \mathrm{LIM}(b_n)$.

(1) *(Embedding the rationals)* For any $q \in \mathbb{Q}$, consider the sequence $(c_n)$ satisfying $c_n = q$ for all $n \in \mathbb{N}$ and let $\mathrm{LIM}(c_n) = q$.
(2) *(Defining addition)* We define the *sum* $x + y$ as $x + y = \mathrm{LIM}(a_n + b_n)$.
(3) *(Defining negatives)* We define $-x = \mathrm{LIM}(-a_n)$.
(4) *(Defining multiplication)* We define the *product* $xy$ as $xy = \mathrm{LIM}(a_n b_n)$.

[1] In this section, we have closely followed the exposition of T. Tao, *Analysis. I.*, Texts Read. Math., 37 Hindustan Book Agency, New Delhi, 2006. xviii+388+XIV pp.
[2] The reader can find all the proofs and details in Chapter 5 of Tao's book.

(5) *(Defining reciprocals)* If $x \neq 0$, then there exists some $c \in \mathbb{Q}$ with $c > 0$ and a $\mathbb{Q}$–Cauchy sequence $(a_n)$ such that $|a_n| \geqslant c$ for all $n \in \mathbb{N}$ which satisfy $x = \mathrm{LIM}(a_n)$.

Moreover, we can further prove that $(a_n^{-1})$ is $\mathbb{Q}$–Cauchy. Thus, we define the *reciprocal* $x^{-1}$ as $x^{-1} = \mathrm{LIM}(a_n^{-1})$.

You can check that all the above definitions are well–defined and moreover that addition and multiplication of these new objects satisfy the standard properties of addition and multiplication (such as commutativity, associativity, et cetera).

**Definition 5.6.** Any *real number* is defined to be $\mathrm{LIM}(a_n)$ for some $\mathbb{Q}$–Cauchy sequence $(a_n)$. We write $\mathbb{R}$ to be the set of all real numbers.

Thus, we now understand how to perform basic algebraic operations with real numbers. We can also order real numbers.

**Definition 5.7.** We say that a real number $x$ is *positive* if there exists some $c \in \mathbb{Q}$ with $c > 0$ and a $\mathbb{Q}$–Cauchy sequence $(a_n)$ such that $a_n \geqslant c$ for all $n \in \mathbb{N}$ and $x = \mathrm{LIM}(a_n)$. We write this as $x > 0$. We can similarly define *negative* real numbers.

Given real numbers $x, y$, we say that $x > y$ if $x - y > 0$.

*Remark* 5.8. One can check that this matches with the standard properties of order on rational numbers. This includes the nice property of order that given any real number $x$, we have either $x = 0$ or $x > 0$ or $x < 0$. Moreover, if $x, y > 0$, then $x + y > 0$ and $xy > 0$.

We now arrive at the first Archimedean type property of real numbers.

**Lemma 5.9.** *Let $x$ be real number such that $x > 0$. Then there exists $q \in \mathbb{Q}$ with $q > 0$ and some $n \in \mathbb{N}$ such that $q \leqslant x \leqslant n$.*

Now that we have a notion of order, we can define what it means for a non-empty set to have an upper bound or a lower bound as in §1.2, as well as what it means for a set to have a least upper bound or a greatest lower bound as in §1.3. Our construction of real numbers can now be used to justify the following familiar property of least upper bounds.

**Lemma 5.10** (Least upper bound property revisited)**.** *Let $E \subset \mathbb{R}$ be a non-empty set such that $E$ has an upper bound. Then $E$ must have a least upper bound.*

5.1. **Arithmetic and order in $\mathbb{R}$** *(Non-examinable).* Using the above construction and lemmas, one can (rather painstakingly) prove the following properties of arithmetic and order in $\mathbb{R}$.

**Addition**

(A1) if $x, y \in \mathbb{R}$ then $x + y \in \mathbb{R}$ (closure under addition)

(A2) if $x, y, z \in \mathbb{R}$ then $(x + y) + z = x + (y + z)$ (addition is associative)

(A3) if $x, y \in \mathbb{R}$ then $x + y = y + x$ (addition is commutative)

(A4) there exists $0 \in \mathbb{R}$ such that for any $x \in \mathbb{R}$, $x + 0 = 0 + x = x$ (existence of an additive identity)

(A5) if $x \in \mathbb{R}$ there exists $-x \in \mathbb{R}$ such that $x + (-x) = (-x) + x = 0$ (existence of an additive inverse)

**Multiplication**

(M1) if $x, y \in \mathbb{R}$ then $xy \in \mathbb{R}$ (closure under multiplication)

(M2) if $x, y, z \in \mathbb{R}$ then $(xy)z = x(yz)$ (multiplication is associative)

(M3) if $x, y \in \mathbb{R}$ then $xy = yx$ (multiplication is commutative)

(M4) there exists $1 \in \mathbb{R}$, such that for any $x \in \mathbb{R}$, $1.x = x.1 = x$ (existence of multiplicative identity)

(M5) if $x \in \mathbb{R}$ with $x \neq 0$, there exists $x^{-1} \in \mathbb{R}$ such that $x.x^{-1} = x^{-1}.x = 1$ (existence of multiplicative inverses)

**Interaction between addition and multiplication**

(D) if $x, y, z \in \mathbb{R}$, $x(y + z) = xy + xz$ (distributive property)

**Order properties**. We use $y > x$ as an equivalent for $x < y$, and $x \leq y$ means that either $x < y$ or $x = y$.

(O1) if $x, y \in \mathbb{R}$ exactly one of the statements $x < y$, $x = y$, $x > y$ is true (trichotomy)

(O2) if $x, y, z \in \mathbb{R}$ then if $x < y$ and $y < z$, $x < z$ (transitivity)

**Interaction between order and arithmetic**

(O3) if $x, y, z \in \mathbb{R}$ and $x < y$, then $x + z < y + z$

(O4) if $x, y, z \in \mathbb{R}$, $x < y$, and $z > 0$, then $xz < yz$

There are some familiar properties of arithmetic that seem to be missing from the above list, but you can prove them using these axioms and some ingenuity. These sort of arguments are a bit tricky, and either fun or painful depending on your taste. It's very rare to have to argue in this sort of 'micro-logical' way.

- there is only one element $0 \in \mathbb{R}$.

  Suppose that there are two elements, 0 and $0'$, that both satisfy (A4). Then $0 = 0 + 0' = 0'$ (first use (A4) for $0'$ and then use (A4) for 0).

- if $x, y, z \in \mathbb{R}$, $x + z = y + z$, then $x = y$. [Additive cancellation.]

  Add $-z$ to both sides, use (A2), then (A5), then (A4):

$$x = x + 0 = x + (z + (-z)) = (x + z) + (-z) = (y + z) + (-z) = y + (z + (-z)) = y + 0 = y.$$

- for any $x \in \mathbb{R}$, $0 \cdot x = 0$.
- for any $x \in \mathbb{R}$, $-x = (-1) \cdot x$.

  From the previous fact and (A5)

$$0 = (1 + (-1))x = x + (-1)x;$$

  use (A5) again and add $-x$ to both sides to give (using (A2) and (A5)) $-x = (-1)x$.
- If $ab = ac$ and $a \neq 0$ then $b = c$. [Multiplicative cancellation.]

  We use (M5), (M2), and (M4):

$$b = 1 \cdot b = (a^{-1} \cdot a)b = a^{-1}(ab) = a^{-1}(ac) = (a^{-1}a)c = 1 \cdot c = c.$$

- $(-a)b = -(ab)$.
- $(-a)(-b) = ab$. Use the previous fact along with $-(-x) = x$.
- if $x \in \mathbb{R}$ with $x > 0$ then $-x < 0$.

  Start with $x > 0$, and add $-x$ to both sides: using (A5) and (O3) gives $0 = (-x) + x > -x$, and $0 > -x$ is the same as $-x < 0$.
- if $x \in \mathbb{R}$ with $x < 0$ then $-x > 0$. (Essentially the same argument.)
- if $x < y$ and $z < 0$ then $xz > yz$.

  Since $-z > 0$, we have $(-z)x < (-z)y$, which is $-zx < -zy$. Now add $zx + zy$ to both sides to obtain $zy < zx$.
- if $0 < a < b$ and $0 < c < d$ then $ac < bd$.

  Since $c > 0$ and $a < b$ we have $ac < bc$ (O4). Since $b > 0$ and $c < d$ we have $bc < bd$ (O4) again. Now we use (O2) [transitivity]: $ac < bc$ and $bc < bd$ implies that $ac < bd$.
- $x^2 > 0$ for every $x \neq 0$.